

REVIEW

A survey and comparison of several friction force models for dynamic analysis of multibody mechanical systems

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Abstract This study is aimed at examining and comparing several friction force models dealing with different friction phenomena in the context of multibody system dynamics. For this purpose, a comprehensive review of present literature in this field of investigation is first presented. In this process, the main aspects related to friction are discussed, with particular emphasis on the pure dry sliding friction, stick–slip effect, viscous friction and Stribeck effect. In a simple and general way, the friction force models can be classified into two main groups, namely the static friction approaches and the dynamic friction models. The former group mainly describes the steady-state behavior of friction force, while the latter allows capturing more properties by using extra state variables. In the present study, a total of 21 different friction force models are described and their fundamental physical and computational characteristics are discussed and compared in details. The application of those friction models in multibody system dynamic modeling and simulation is then investigated. Two multibody mechanical systems are utilized as demonstrative application examples with the purpose of illustrating the influence of the various frictional approaches on the dynamic response of the

systems. From the results obtained, it can be stated that both the choice of the friction force model and friction parameters involved can significantly affect the simulated/modeled dynamic response of mechanical systems with friction.

Keywords Dry contact · Friction force laws · Static models · Dynamic models · Multibody dynamics

1 Introduction

It is known that friction forces of complex nature can be found in all actual mechanical systems that have contacting surfaces with relative motion. In most of the cases, these forces are not desirable due to their dissipative effect and wear generation tendencies, while sometimes they are demanded for the correct operation of the systems. However, both situations require a rigorous evaluation of friction forces in order to obtain an accurate dynamic response of the system. Thus, the objective of this work is twofold: first to present a critical overview on the main static and dynamic friction force models and second to investigate the influence of friction modeling on the dynamic behavior of multibody mechanical systems.

By and large, friction can be associated with the resistance to the relative motion between two surfaces in contact. Friction is a highly complex phenomenon, and due to its importance, it has been studied over the centuries by many researchers. One of the first

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works appeared in the sixteenth century resulting from Leonardo da Vinci's studies, who stated that the friction force is proportional to normal force, opposes to the motion and is independent of the contact area. Later, Amontons [1] corroborated with da Vinci on his work. Coulomb [2] extended their work by stating that friction is independent of velocity magnitude, and developed the first mathematical friction model. Indeed, it can be said that the Coulomb friction law was the predecessor for all of the friction models. This approach is quite simple and involves a single parameter, that is, the coefficient of friction. However, the dry Coulomb's friction approach fails to describe most of the frictional effects and presents a discontinuity for the friction force at zero velocity since the force is dependent on the direction of the relative velocity. This discontinuity is an important issue for dynamic modeling and simulations because it introduces numerical instability in the system's response.

Further researches on frictional behavior suggested the existence of higher friction forces at rest [3–5], which led to considering two different friction coefficients, one for “static” friction and another for “kinetic” friction. Bearing that in mind, in case that the relative velocity between two surfaces vanishes, the friction force will increase which leads to the sticking thereof. This phenomenon is the so-called stick–slip [6, 7], and it is overcome since the external tangential force is larger than the static friction force, so the bodies start sliding. Several researchers have experimentally verified the influence of this phenomenon on the mechanical system's response [8–10]. The stick–slip is a major problem during the modeling and simulation of mechanical systems, since it is an unstable phenomenon and not easy to predict. Furthermore, it is important to choose a friction model capable of capturing its behavior and its influence on the system's dynamics. In this sense, the model has to consider different values for the coefficient of friction for the static and kinetic cases, but most of all it should describe a correct transition between them. This transition was studied by Stribeck [11], who experimentally showed that, for low velocities, friction decreases with the increase in the relative velocity. This effect implies a continuous drop of friction force, which contradicts the discontinuous behavior of considering just two different coefficients of friction. This behavior is described by the Stribeck curve, which was named after him, and represents the steady-state friction as a function of velocity. Rabinow-

icz [4] studied the essence of the static and the kinetic coefficients of friction and their transition. The experimental results showed that the coefficient of friction is a function not only of the relative velocity, but also of the displacement. It was demonstrated that the maximum friction force is reached for low displacements. Some other studies [12, 13] have been made in order to examine the influence of displacement in friction forces. It has been shown experimentally that when any external tangential force is applied between two contacting bodies, a displacement always occurs, which is thought of as a result of surface elasticity. This phenomenon of a small motion in elastic range when the applied force is less than the breakaway force is often called pre-sliding displacement. The explanation for this small displacement relies on the elastic deformation of the asperities of the contacting surfaces [14]. The “adhesion theory” states that in the presence of a normal load the asperities tend to adhere. Hence, when a tangential force is applied, the bond tends to break. Before that happening, the asperities should have elastic, and then plastic deformation, which causes the relative displacement of the contacting surfaces. Since the junctions of asperities are broken, the sticking regime ends and the bodies start sliding. The necessary force to initiate the motion and surmount the static friction is called the breakaway force, which is also the maximum friction force [4]. Normally, the overcoming of this force defines the boundary between the sticking and sliding regimes. Johannes et al. [15] demonstrated that the breakaway force depends on the rate of the external applied force. It was shown that the increase in force rate leads to the reduction in the breakaway force. The nonlinear behavior of frictional phenomena can be related to the appearance of self-excited vibrations which provoke a chaotic motion of the contacting bodies [16–18]. These vibrations are mainly related to the Stribeck effect, since the velocity dependence produces changes in the friction force, which can lead to sudden changes in the state of the system. Some researchers have been studying the effects of these vibrations and how to accurately model the transitions between stick and slip [19–22]. Further investigations focused on the stable behavior of frictional vibrations which may undergo on bifurcations [17, 18, 23].

The existence of a fluid lubricant layer between two rubbing surfaces has a significant influence in the resultant friction force. Thus, this needs to be taken into account when using a friction model [24]. To do so,

a linear relation between the relative velocity and the viscous friction force is generally employed, because most of fluid lubricants can be considered Newtonian fluid. Normally, the component of friction force related to viscosity is simply summed to the regular friction model. Another important phenomenon is the “frictional lag,” which consists of the delay in change in friction force as a function of velocity change rate. This characteristic behaves as an inertia component to the change in friction state. As a consequence, there is a hysteresis loop in the relationship between friction force and velocity. Hess and Soom [25] verified empirically that the friction force does not respond instantaneously to the variation of velocity. Thus, frictional lag is a dynamical behavior, which results in lower friction forces when velocity is decreasing and larger forces for increasing velocities [26]. This phenomenon becomes more evident with larger accelerations and enhances the need to use dynamic friction models. Most of these phenomena are not included into the classic Coulomb’s law; however, it is still widely used due to its simplicity, and for that reason, several friction models have been developed over the years based on modifications of Coulomb’s approach [27–29]. Most of these are aimed to eliminate the discontinuity at zero velocity. The primal friction models are quite simple, although the complexity associated with frictional effects introduces the need to develop more robust models. A significant number of friction models can be found in the literature. In order to perform a comparative analysis, they are usually classified, and the most utilized criterion consists on the separation between “static” and “dynamic” models [29–32]. The static friction models usually describe the steady-state behavior of the friction force and are in general simpler. On the other hand, the dynamic models are aimed to describe friction more realistically due to the inclusion of extra state variables, thus giving an additional degree of freedom for each frictional contact. In addition to these two categories, it is also common to classify the acceleration-based friction models, which as their name suggests, evaluate the friction force as a function of the relative acceleration of the contacting surfaces. In the scope of this work, the latter group will be considered as a portion of the static friction models.

Since the Coulomb friction model is not capable of describing most of friction characteristics, other static models have been suggested. Several researchers have proposed different mathematical expressions to describe the Stribeck curve [25, 33–36], although the

exponential approximation presented by Bo and Pavlescu [37] showed greater acceptance. Karnopp [38] suggested an approach in which the relative velocity is considered to be null within a given velocity range, and the friction force is evaluated as the velocity was zero. This model eliminates the discontinuity at zero velocity, and it is capable of capturing stick–slip motion. In order to overcome some numerical instability of Karnopp model, Leine et al. [20] proposed the “switch model.” Armstrong-Hélouvry et al. [39] presented a survey on friction models and introduced the seven-parameter model which is constituted by two state equations, one for sliding and another for sticking. McMillan [19] studied the self-excited vibrations generated by friction phenomena and introduced a phenomenological model in which the friction force is evaluated based on the relative acceleration as well. More recently, Wojewoda et al. [40] proposed a model, which describes the hysteretic behavior of friction, and Awrejcewicz et al. [41] modeled the friction force as a function of the external tangential force for low velocities.

The dynamic models emerged along with the need of capturing some frictional characteristics that the static models were not able to describe. Dahl [42, 43] introduced the first dynamic model based on the stress–strain curve to model the frictional behavior of ball bearings. Comparing to the Coulomb friction law, both models do not capture stick–slip motion, but it was shown that Dahl model presents better results [44, 45], since this model is capable of describing the pre-sliding displacement. Most of the dynamic friction models are based on the physical interaction between the surfaces asperities, such as the bristle model [46], the reset integrator [46], the LuGre [47], the elasto-plastic [48], among others [49–56]. Generally, these approaches consider an extra state variable related to the bristles deflection.

The referred models are typically suitable for describing the friction force at a point contact; therefore, they can successfully model in a very efficient way the interaction between two surfaces, if the contact area is quite small. However, there are a significant number of scenarios in which the rolling resistance torque generated in the contact cannot be neglected. This phenomenon was first noticed by Contensou [57], who stated that the friction resistance is function of the relative sliding velocity and relative normal angular velocity (spinning). Several authors [58–64] paid attention to this issue by proposing different approaches to determining

the frictional rolling resistance, also called Coulomb–Contensou’s friction. This phenomenon is most noticeable when the resistant torque is most responsible for the motion of the system, as in the Tippe Top [59,61] or the Celtic Stone [65,66].

Regarding the selection of the friction force model to adopt for the simulation of a mechanical system, the easiness of establishing the parameters to be used for the simulation is of crucial importance. Dahl [67] presented an experimental methodology to determine the parameters of the friction model [42]. The selection of the static and kinetic friction coefficients was addressed by Borsotto et al. [68]. The selection of parameters for Karnopp friction model has also been examined [69,70]. Wu et al. [71] introduced an experimental procedure for the identification of the parameters of LuGre model to simulate tire friction. Piatkowski [72] studied parameters selection for the Dahl and LuGre models and concluded that a certain set of parameters is only valid for a specific range of normal contact force. Lately, Sun et al. [73] established a new experimental setup for parameters identification that consists of a mass block connected to a ball screw mechanism sliding over a metal base.

Due to its paramount importance and deep interest from many researchers, several surveys on friction characteristics and their implementation in mechanical systems can be found in the literature [29,30,39,74,75]. Awrejcewicz and Olejnik [74] presented a review on different systems with impact and friction, which show nonlinear behavior, and special attention was paid to some complex bifurcations and chaotic motion. Recently, Pennestrì et al. [75] assessed the impact of some of the most relevant friction force models on the computational efficiency of the dynamic analysis of simple mechanical systems. In the context of multibody dynamics, some researchers have been studying the modeling and implementation of friction into kinematic joints [76–80]. Piedbœuf and his coauthors [79] proposed a numerical algorithm for modeling friction in robotic manipulators with revolute joints. In a different manner, Fręgonaczek and Wojtyra [80] studied uniqueness of direct dynamics simulation of rigid multibody system with redundant constraints and frictional joints.

The experimental validation of friction modeling is of paramount importance to ensure the proficiency to model the several phenomena associated with friction. Lampaert et al. [81] provided a comparison between

a static model with Stribeck curve, Dahl, LuGre and Leuven, and they showed that Leuven model has a better agreement with experimental results. Tjahjowidodo et al. [82] used a DC motor to provide a comparison between the Coulomb, Stribeck static friction, LuGre and GMS approaches, and the results presented a better fit for GMS model. They also saw that for high velocities, the Coulomb and Stribeck models also provided good results. Liang et al. [55] used two experimental test rigs to validate the friction model proposed. More recently, Liu et al. [83] performed a study where the dynamic behaviors modeled by the Coulomb, Stribeck, Dahl, LuGre and elasto-plastic approaches have been compared with the experimental measurements, and the results showed that the LuGre approach presented a closer behavior compared to experiment.

Thus, this work is aimed to present and compare several friction force models that can be utilized in the context of multibody systems formulations. In the sequel of this process, the most relevant static and dynamic friction models are briefly described. Subsequently, two applications examples are taken into account to illustrate the dynamic response of multibody systems using different friction approaches. The computational efficiency and the impact of their parameters are also assessed in the discussion of results. Finally, the main conclusions from this study are outlined in terms of selection of proper friction models, based on which the engineering analysis and design implications of the outcomes can be addressed.

2 Mathematical description of “static” friction force models

This section includes several “static” friction force models frequently used in the simulations of multibody mechanical systems. It must be stated that most of these models present a discontinuity of friction force when the relative velocity is zero, which can cause difficulties in describing friction realistically. In general, these models have some limitations as regards reproducing the previously mentioned frictional phenomena.

2.1 Coulomb friction

Coulomb [2] presented the first friction model, which states that the friction always opposes relative motion between contacting bodies and its magnitude is proportional to the normal contact force. This model depends

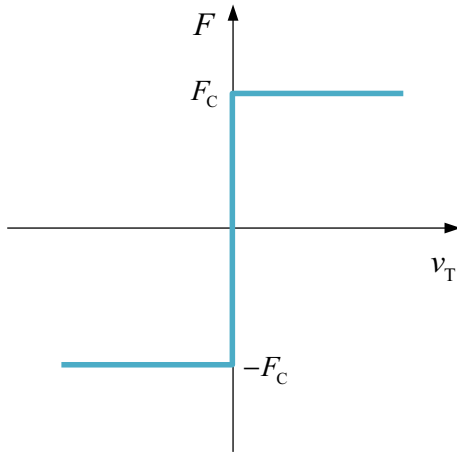


Fig. 1 Representation of Coulomb friction for 1D case

on the relative velocity direction, except for zero velocity where the friction force is a multivalued function of the external tangential force. This model can be described as

$$\mathbf{F} = \begin{cases} F_C \operatorname{sgn}(\mathbf{v}_T) & \text{if } \|\mathbf{v}_T\| \neq 0 \\ \min(\|\mathbf{F}_e\|, F_C) \operatorname{sgn}(\mathbf{F}_e) & \text{if } \|\mathbf{v}_T\| = 0 \end{cases} \quad (1)$$

where

$$F_C = \mu_k \|\mathbf{F}_N\| \quad (2)$$

in which $\mathbf{F}_N$ is the normal contact force, F_C denotes the magnitude of Coulomb friction, μ_k is the kinetic coefficient of friction, $\mathbf{F}_e$ represents the resultant of the external forces acting on the reference body in the tangential direction of the contact, $\mathbf{v}_T$ is the relative tangential velocity of the body with respect to the other contacting surface and $\mathbf{F}$ denotes the friction force that the reference body exerts in the contacting surface. This model presents a dependence on the velocity by the *signum* function,

$$\operatorname{sgn}(\mathbf{v}_T) = \begin{cases} \frac{\mathbf{v}_T}{\|\mathbf{v}_T\|} & \text{if } \|\mathbf{v}_T\| \neq 0 \\ \mathbf{0} & \text{if } \|\mathbf{v}_T\| = 0 \end{cases} \quad (3)$$

where $\mathbf{0}$ is a null vector with the same dimensions as $\mathbf{v}_T$. In Fig. 1, the friction force is represented as a function of the relative velocity, for the 1D case.

Although this model is straightforward, it presents some difficulties of implementation since it does not specify a friction force at null velocity. Thus, this

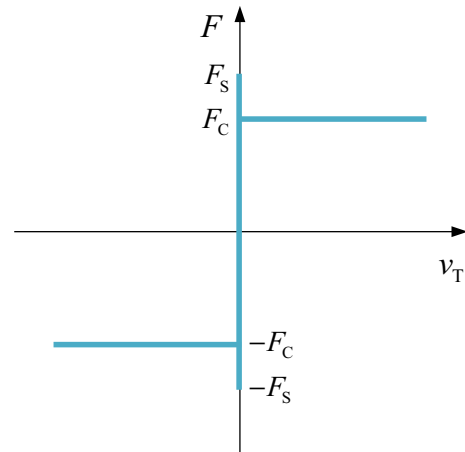


Fig. 2 Representation of the Coulomb model with stiction for 1D case

velocity dependence can originate perturbations in the dynamic simulations of system's response. Nevertheless, the Coulomb friction law is used quite often to simulate friction behavior for the sake of simplicity, since it only requires one input parameter, i.e., the coefficient of friction.

2.2 Coulomb model with stiction

Since the friction force at null velocity is higher than the kinetic friction, several studies have presented the necessity of introducing a friction model which includes two different friction coefficients. This modified Coulomb approach has a similar behavior to Coulomb's except in the vicinity of zero velocity. It is also a multivalued function, but can reach a higher friction force, and can be described as follows

$$\mathbf{F} = \begin{cases} F_C \operatorname{sgn}(\mathbf{v}_T) & \text{if } \|\mathbf{v}_T\| \neq 0 \\ \min(\|\mathbf{F}_e\|, F_S) \operatorname{sgn}(\mathbf{F}_e) & \text{if } \|\mathbf{v}_T\| = 0 \end{cases} \quad (4)$$

where

$$F_S = \mu_s \|\mathbf{F}_N\| \quad (5)$$

in which F_S is the magnitude of static friction and μ_s denotes the static coefficient of friction which is higher than the kinetic one μ_k . The variation of the friction force with the relative velocity is shown in Fig. 2.

Although this model considers stiction, since it hardly reaches exact null velocity during a simulation, it in practice provides similar behavior compared to

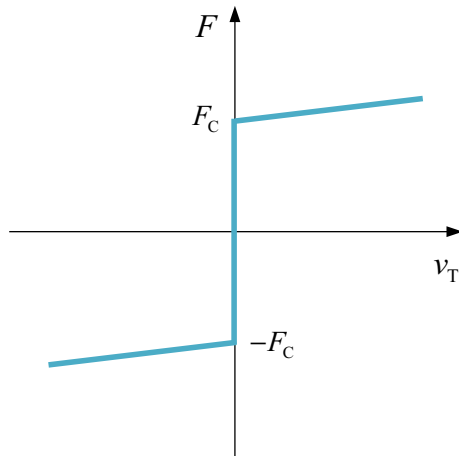


Fig. 3 Representation of Coulomb model with linear viscous friction for 1D case

Coulomb's law with an oscillatory force for low velocities.

2.3 Coulomb model with viscous friction

One of most common modification of Coulomb's friction law is the introduction of a viscous friction component. Considering a linear relationship between the relative velocity and the friction force related to the lubricants viscosity, the friction model can be written as

$$\mathbf{F} = \begin{cases} F_C \operatorname{sgn}(\mathbf{v}_T) + F_v \mathbf{v}_T & \text{if } \|\mathbf{v}_T\| \neq 0 \\ \min(\|\mathbf{F}_e\|, F_C) \operatorname{sgn}(\mathbf{F}_e) & \text{if } \|\mathbf{v}_T\| = 0 \end{cases} \quad (6)$$

where F_v is the viscous friction coefficient which is deeply related to the viscosity of the fluid. The behavior of this model is presented in Fig. 3.

However, sometimes a nonlinear relation with the velocity can be adopted, because it could be a better fit compared with the experimental data. Notwithstanding, this type of approximation is not commonly employed, since it requires an extra parameter to define the degree of nonlinearity.

2.4 Model with Stribeck effect

In contrast to the Coulomb model with stiction, the Stribeck effect [11] ensures that the decrease from static to kinetic friction is a continuous process. Thus, the friction force during relative motion is expressed as a continuous function of velocity as

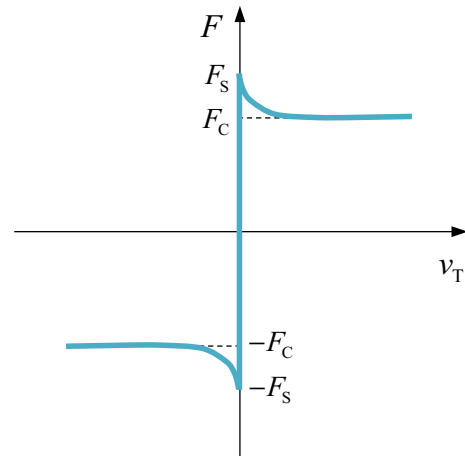


Fig. 4 Representation of the Stribeck curve without viscous friction

$$\mathbf{F} = \begin{cases} \mathbf{F}(\mathbf{v}_T) & \text{if } \|\mathbf{v}_T\| \neq 0 \\ \min(\|\mathbf{F}_e\|, F_S) \operatorname{sgn}(\mathbf{F}_e) & \text{if } \|\mathbf{v}_T\| = 0 \end{cases} \quad (7)$$

where $\mathbf{F}(\mathbf{v}_T)$ is an arbitrary function that depends on the relative velocity. With this model, the friction force diminishes when the relative motion is initiated, as represented in Fig. 4.

Several mathematical expressions have been proposed to describe this curve. Hess and Soom [27] suggested

$$\mathbf{F}(\mathbf{v}_T) = \left(F_C + \frac{F_S - F_C}{1 + \left(\frac{\|\mathbf{v}_T\|}{v_S} \right)^2} \right) \operatorname{sgn}(\mathbf{v}_T) + F_v \mathbf{v}_T \quad (8)$$

where v_S denotes the Stribeck velocity. However, the most popular approach was introduced by Bo and Pavlescu [37], which considers an exponential function as

$$\mathbf{F}(\mathbf{v}_T) = \left(F_C + (F_S - F_C) e^{-\left(\frac{\|\mathbf{v}_T\|}{v_S} \right)^{\delta_\sigma}} \right) \operatorname{sgn}(\mathbf{v}_T) + F_v \mathbf{v}_T \quad (9)$$

in which δ_σ is a factor that relies on the geometry of the contacting surfaces, often considered to be equal to 2 [35].

This friction model takes into account the Coulomb, viscous, stiction and Stribeck friction effects. However, it presents the same problem as the previous approaches at null velocity.

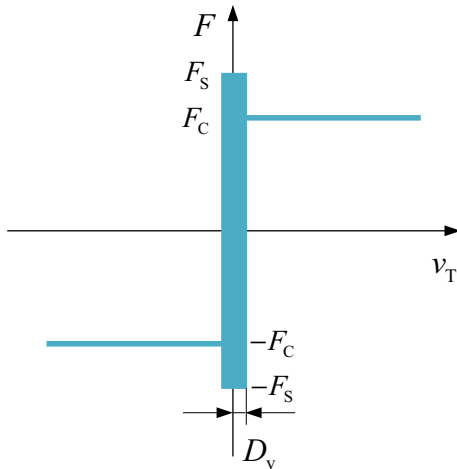


Fig. 5 Representation of Karnopp model with Coulomb friction for 1D case

2.5 Karnopp model

Since the aforementioned models are multivalued functions for zero velocity, their static behavior cannot be captured during a simulation. To overcome this difficulty, Karnopp [38] proposed a model where the velocity is considered zero, for a specified range, as shown in Fig. 5. Thus, when the velocity is within this interval, the system's state can change and the model's response will be the same as when the relative velocity is zero.

Karnopp model is usually used along with the Coulomb model and can be expressed as

$$\mathbf{F} = \begin{cases} \mathbf{F}(\mathbf{v}_T) & \text{if } \|\mathbf{v}_T\| > D_v \\ \min(\|\mathbf{F}_e\|, F_S) \operatorname{sgn}(\mathbf{F}_e) & \text{if } \|\mathbf{v}_T\| \leq D_v \end{cases} \quad (10)$$

where D_v is the tolerance for zero velocity and $\mathbf{F}(\mathbf{v}_T)$ is a given function for the friction force outside of the tolerance velocity. It is important to select a suitable range of the null velocity [69,70]. Nevertheless, this zero velocity interval does not comply with the real behavior of the contact.

2.6 Switch model

In order to reduce the numerical instability for the adhesion phase, Leine et al. [20] proposed a modification of Karnopp friction model. The main difference between both models occurs during the adhesion phase when the external tangential force is lower than the maximum static friction force. In that case, for Karnopp

model, the friction force equalizes the external tangential force, which means there is null acceleration in the tangential direction.

However, having a null acceleration with a non-null velocity results in numerical instability. To solve that problem, Leine et al. suggested the direct determination of the acceleration $\mathbf{a}$ instead of calculating the friction force, as

$$\mathbf{a} = -\mathbf{v}_T \sqrt{\frac{k}{m}} \quad (11)$$

where k is the stiffness of the external force and m is the mass of the body. Thus, the acceleration is exactly zero only for null relative velocity.

2.7 Models with finite slope at zero velocity

In most of the static models described earlier, the friction force at zero velocity is multivalued and evaluated as a function of the external tangential force. In order to simplify and ensure computational efficiency, several researchers have proposed alternative methods, which replace the discontinuity at zero velocity by a finite slope model. This strategy normally avoids computational instability due to the change in the direction of the velocity.

In order to smooth out the discontinuity of the Coulomb friction model at zero velocity, several modifications were proposed [29]. One approach consists of using linear friction-velocity dependence for low velocities, which can be expressed by the following equation

$$\mathbf{F} = F_C \min(k \|\mathbf{v}_T\|, 1) \operatorname{sgn}(\mathbf{v}_T) \quad (12)$$

where k is a coefficient that gives the slope for null velocity. This model is displayed in Fig. 6 for the 1D case.

Another possibility resides in utilizing a hyperbolic tangent approximation, which is numerically a more stable model since it has a continuous derivative. The friction force is thus given by

$$\mathbf{F} = F_C \tanh(k \|\mathbf{v}_T\|) \operatorname{sgn}(\mathbf{v}_T) \quad (13)$$

Threlfall [27] also presented a model, which softens the existing discontinuity in the Coulomb law. The curve

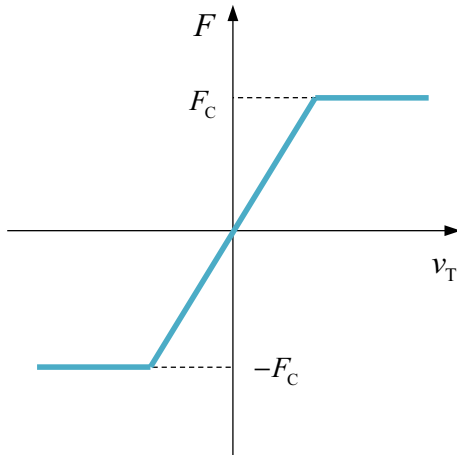


Fig. 6 Representation of Coulomb model with linear friction modification for 1D case

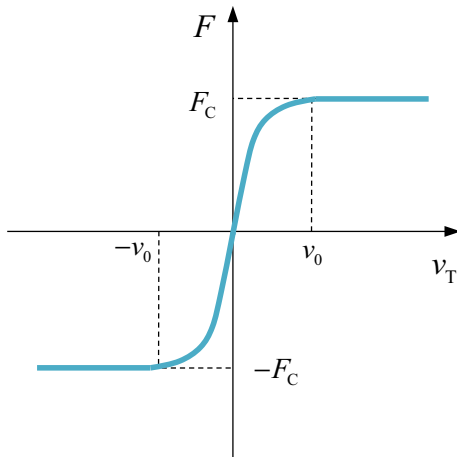


Fig. 7 Representation of Threlfall model for 1D case

that represents this model is shown in Fig. 7, and it can be described by the following expression

$$\mathbf{F} = \begin{cases} F_C \left(1 - e^{-\frac{3\|\mathbf{v}_T\|}{v_0}} \right) \text{sgn}(\mathbf{v}_T) & \text{if } \|\mathbf{v}_T\| \leq v_0 \\ 0.95 F_C \text{sgn}(\mathbf{v}_T) & \text{if } \|\mathbf{v}_T\| > v_0 \end{cases} \quad (14)$$

where v_0 is a tolerance velocity. For high velocities, Eq. (14) returns a friction force slightly lower than Coulomb's friction. This value was determined in order to avoid a discontinuity in the transition between the two parts of this piecewise function. However, in this present work, a corrected version has been considered, which can be expressed as

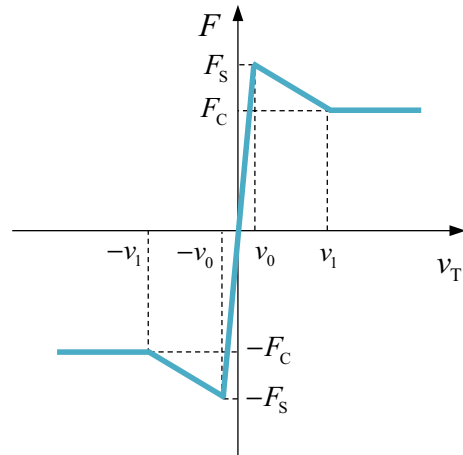


Fig. 8 Representation of the linear model with stiction for 1D case

$$\mathbf{F} = \begin{cases} F_C \left(\frac{1 - e^{-\frac{3\|\mathbf{v}_T\|}{v_0}}}{1 - e^{-3}} \right) \text{sgn}(\mathbf{v}_T) & \text{if } \|\mathbf{v}_T\| \leq v_0 \\ F_C \text{sgn}(\mathbf{v}_T) & \text{if } \|\mathbf{v}_T\| > v_0 \end{cases} \quad (15)$$

The continuity in the transition between both parts has been ensured by this equation. Furthermore, the model has a higher resemblance with Coulomb friction law with the decrease in the tolerance velocity.

The use of a finite slope at zero velocity has been also extended to models, which take into account the stiction. One of the most popular approaches consists in considering a peak of friction force for a given velocity and performing a linear interpolation, as shown in Fig. 8. The model can be expressed as

$$\mathbf{F} = \begin{cases} \left(\frac{\|\mathbf{v}_T\|}{v_0} F_S \right) \text{sgn}(\mathbf{v}_T) & \text{if } \|\mathbf{v}_T\| \leq v_0 \\ \left(F_S - \frac{\|\mathbf{v}_T\| - v_0}{v_1 - v_0} (F_S - F_C) \right) \text{sgn}(\mathbf{v}_T) & \text{if } v_0 < \|\mathbf{v}_T\| < v_1 \\ F_C \text{sgn}(\mathbf{v}_T) & \text{if } \|\mathbf{v}_T\| \geq v_1 \end{cases} \quad (16)$$

where v_0 and v_1 are the tolerances for the tangential velocity.

Bengisu and Akay [84] proposed a model capable of capturing the Stribeck effect, as represented in Fig. 9. The model is constituted by two equations (one for the slope and another to describe the Stribeck effect), which are given by

$$\mathbf{F} = \begin{cases} \left(-\frac{F_S}{v_0^2} (\|\mathbf{v}_T\| - v_0)^2 + F_S \right) \text{sgn}(\mathbf{v}_T) & \text{if } \|\mathbf{v}_T\| < v_0 \\ \left(F_C + (F_S - F_C) e^{-\xi(\|\mathbf{v}_T\| - v_0)} \right) \text{sgn}(\mathbf{v}_T) & \text{if } \|\mathbf{v}_T\| \geq v_0 \end{cases} \quad (17)$$

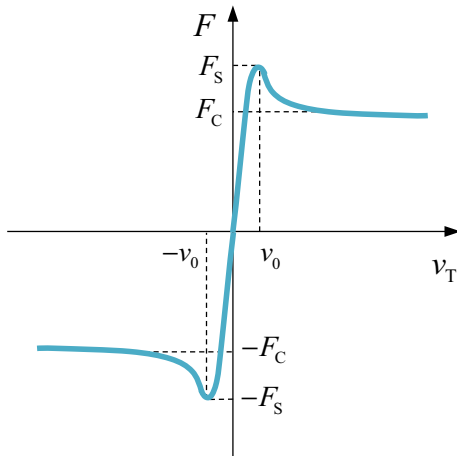


Fig. 9 Representation of the Bengisu and Akay model for 1D case

in which ξ should be a positive parameter which represents the negative slope of the sliding state.

However, these models have some particularities. When the slope at zero velocity is too large, a small time step increment is needed to correctly capture friction for low velocities, which will slow down the simulation. In addition, for velocities close to zero, the friction force will always be low independently of the displacement.

2.8 Ambrósio model

The above-mentioned limitations associated with the discontinuity of the friction force led Ambrósio [28] to propose a modified Coulomb's friction law in which the friction force can be defined as

$$\mathbf{F} = \begin{cases} \mathbf{0} & \text{if } \|\mathbf{v}_T\| \leq v_0 \\ \frac{\|\mathbf{v}_T\| - v_0}{v_1 - v_0} F_C \text{sgn}(\mathbf{v}_T) & \text{if } v_0 < \|\mathbf{v}_T\| < v_1 \\ F_C \text{sgn}(\mathbf{v}_T) & \text{if } \|\mathbf{v}_T\| \geq v_1 \end{cases} \quad (18)$$

where v_0 and v_1 are the tolerances for the velocity. The main difference of this model is the null friction force for low velocities, as shown in Fig. 10.

This approach prevents the friction force from changing direction when the relative velocity is close to zero. However, it does not describe the stick-slip motion.

2.9 The seven-parameter model

This model is also known as Armstrong model [39], and it includes several frictional effects such as the

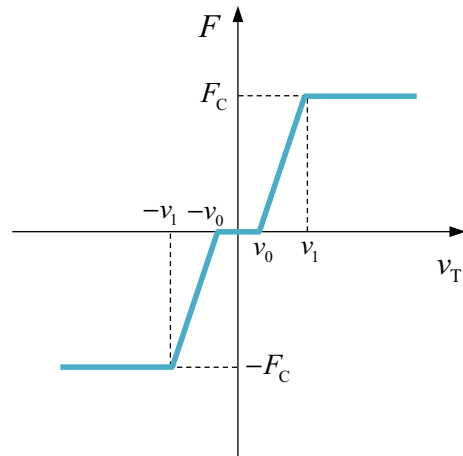


Fig. 10 Representation of Ambrósio friction model for 1D case

pre-sliding displacement, Coulomb and viscous friction, Stribeck effect and frictional lag. It consists of two separate equations, one for adhesion

$$\mathbf{F} = \sigma_0 \mathbf{x} \quad (19)$$

and another for the sliding regime

$$\mathbf{F} = \left(F_C + \frac{F_S(\gamma, t_d) - F_C}{1 + \left(\frac{\|\mathbf{v}_T(t - \tau_L)\|}{v_S} \right)^2} \right) \text{sgn}(\mathbf{v}_T) + F_v \mathbf{v}_T \quad (20)$$

with

$$F_S(\gamma, t_d) = F_{S,a} + (F_{S,\infty} - F_{S,a}) \frac{t_d}{t_d + \gamma} \quad (21)$$

where F_C is the Coulomb friction force, F_v denotes the viscous friction coefficient, F_S is the magnitude of static friction, $F_{S,a}$ represents the magnitude of static friction at the end of the previous sliding period, $F_{S,\infty}$ is the magnitude of static friction after a long time at rest, σ_0 expresses the tangential stiffness of the contact, v_S is the Stribeck velocity, τ_L denotes the time constant of frictional memory, γ is the temporal parameter of the rising static friction and t_d is the dwell time, i.e., the time at zero velocity.

Although this approach seems to describe most of the characteristics of friction behavior, it also has some drawbacks. Firstly, to implement it is necessary to define a large number of parameters. Secondly, the

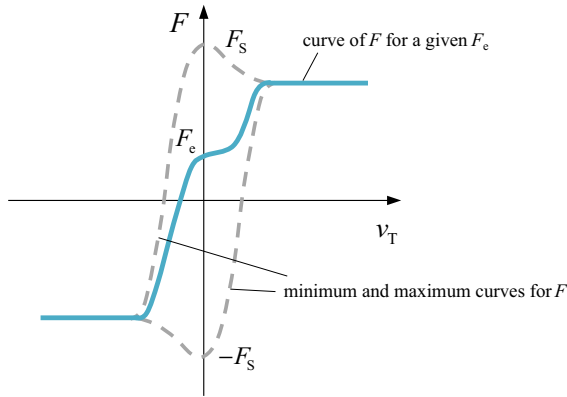


Fig. 11 Representation of the Awrejcewicz et al. model for 1D case

Armstrong model is constituted by two separate states, and there is no criterion defined to determine when the switch must occur. Therefore, this must be defined an eighth parameter to evaluate if the model is to switch between states.

2.10 Awrejcewicz et al. model

Awrejcewicz et al. [41] proposed a static friction model for dry contact, which is dependent of both tangential force and relative velocity. This model is governed by four different equations, one for sliding, two for the transition from stick to slip and one for sticking, as follows

$$\mathbf{F} = \begin{cases} \mathbf{F}(\mathbf{v}_T) & \text{if } \|\mathbf{v}_T\| > v_0 \\ F_S \text{sgn}(\mathbf{F}_e) & \text{if } \|\mathbf{v}_T\| \leq v_0 \wedge \|\mathbf{F}_e\| > F_S \wedge \mathbf{F}_e \cdot \mathbf{v}_T \geq 0 \\ (2A - 1) F_S \text{sgn}(\mathbf{v}_T) & \text{if } \|\mathbf{v}_T\| \leq v_0 \wedge \|\mathbf{F}_e\| > F_S \wedge \mathbf{F}_e \cdot \mathbf{v}_T < 0 \\ A(-\mathbf{F}_e + F_S \text{sgn}(\mathbf{v}_T)) + \mathbf{F}_e & \text{if } \|\mathbf{v}_T\| \leq v_0 \wedge \|\mathbf{F}_e\| \leq F_S \end{cases} \quad (22)$$

in which

$$A = \frac{\|\mathbf{v}_T\|^2}{v_0^2} \left(3 - 2 \frac{\|\mathbf{v}_T\|}{v_0} \right)$$

where v_0 is a velocity tolerance, $\mathbf{F}(\mathbf{v}_T)$ is an arbitrary friction function for sliding which depends on the velocity, and the operator $(\cdot)$ denotes the scalar product of two vectors. The tolerance velocity defines the limit for sliding state. Below this tolerance, the friction force is also calculated as a function of the external tangential force. In Fig. 11, it is displayed a curve of friction force for a given external tangential force (F_e). There

are also represented the curves for the minimum and maximum friction force.

2.11 Wojewoda et al. model

In order to capture most of friction features, Wojewoda et al. [40] proposed a static friction model for dry contact of hysteretic type with a stochastic component. This model is divided into three different states, one for sticking and two for sliding (acceleration and deceleration), and it can be expressed as

$$\mathbf{F}(\mathbf{v}_T, \dot{\mathbf{v}}_T) = \begin{cases} F_{st} \text{sgn}(\mathbf{v}_T) & \text{if } F_{st} \leq F_{d+} \wedge \text{sgn}(\mathbf{v}_T \cdot \dot{\mathbf{v}}_T) \geq 0 \\ F_{d+} \text{sgn}(\mathbf{v}_T) & \text{if } F_{st} > F_{d+} \wedge \text{sgn}(\mathbf{v}_T \cdot \dot{\mathbf{v}}_T) \geq 0 \\ F_{d-} \text{sgn}(\mathbf{v}_T) & \text{if } \text{sgn}(\mathbf{v}_T \cdot \dot{\mathbf{v}}_T) < 0 \end{cases} \quad (23)$$

with

$$F_{st} = \frac{1}{2} k_S \frac{\|\mathbf{v}_T\|^2}{\|\dot{\mathbf{v}}_T\|} - F_0 \quad (24)$$

$$F_{d+} = F_C + (F_S(\mathbf{v}_T) - F_C)(g(\mathbf{v}_T, \dot{\mathbf{v}}_T) + f_R(\mathbf{x}, \mathbf{v}_T)) \quad (25)$$

$$F_{d-} = F_C - (F_S - F_C)(g(\mathbf{v}_T, \dot{\mathbf{v}}_T) + f_R(\mathbf{x}, \mathbf{v}_T)) \quad (26)$$

where F_{st} is the friction for the sticking phase, F_{d+} and F_{d-} represent the friction force in acceleration and deceleration, respectively, k_S is the contact stiffness, F_0 denotes the initial value for sticking force, $f_R(\mathbf{x}, \mathbf{v}_T)$ is

a stochastic function and $g(\mathbf{v}_T, \dot{\mathbf{v}}_T)$ expresses a function to model the Stribeck curve. The stochastic function intends to simulate random features as variations of normal force or inhomogeneous asperity of contacting surfaces.

The researchers considered a variable static friction force during the acceleration phase, as it is shown in Eq. (25). They suggested the following expression

$$F_S(\mathbf{v}_T) = F_S + \Delta F_S \frac{1}{1 + \frac{\|\mathbf{v}_T\|}{v_S}} \quad (27)$$

where ΔF_S represents the range of breakaway force variation and v_S is the Stribeck velocity.

3 Mathematical description of “dynamic” friction force models

This section includes some of the most relevant “dynamic” friction models. As described earlier, in general, the static friction approaches have limitations in capturing some friction phenomena, such as pre-sliding displacement or frictional lag. Thus, better alternatives should be discussed, namely the available dynamic friction models, also named “state variable” models. In a simple manner, the dynamic models introduce extra state variables, used together with the velocity for the evaluation of the friction force.

3.1 Dahl model

The Dahl friction model [42,43] was developed with the aim of describing the friction behavior of ball bearings. The basis of this solution is an analogy with the classical stress–strain curve of materials. Dahl observed that in brittle materials, the difference between the stiction and Coulomb friction is difficult to capture. Ductile materials, however, have more probably the stiction behavior and then decrease the stress until Coulomb friction is reached. Moreover, it was shown that the friction force is dependent on relative velocity and displacement. Dahl model states that when the contacting surfaces are subjected to stress, the friction force increases until rupture of the bonds occurs. In this context, the stress–strain curve can be described by a differential equation as

$$\frac{dF}{dx} = \sigma_0 \left| 1 - \frac{F}{F_C} \operatorname{sgn}(v_T) \right|^\alpha \operatorname{sgn} \left(1 - \frac{F}{F_C} \operatorname{sgn}(v_T) \right) \quad (28)$$

where F denotes the friction force, x is the displacement, F_C is the Coulomb friction, σ_0 represents the stiffness coefficient and α is a parameter that defines the shape of the material curve. This parameter depends on the material, usually varies between 0 and 1 for brittle materials and is higher than 1 for ductile materials. From the analysis of Eq. (28), when F tends to F_C , the

derivative tends to zero, so it can be concluded that the magnitude of the friction force does not exceed F_C .

Equation (28) can be modified into a time derivative, and this formulation can be generalized for the 3D case. Furthermore, the most common value for α is 1, yielding

$$\frac{d\mathbf{F}}{dt} = \sigma_0 \left(1 - \frac{\mathbf{F} \cdot \operatorname{sgn}(\mathbf{v}_T)}{F_C} \right) \mathbf{v}_T \quad (29)$$

Introducing the state variable $\mathbf{z}$, and assuming that $\mathbf{F} = \sigma_0 \mathbf{z}$, Eq. (29) can be rewritten as

$$\frac{d\mathbf{z}}{dt} = \left(1 - \frac{\sigma_0}{F_C} \mathbf{z} \cdot \operatorname{sgn}(\mathbf{v}_T) \right) \mathbf{v}_T \quad (30)$$

It can be observed from Eq. (30) that when the system reaches the steady state, the friction force can be simplified as

$$\mathbf{F} = F_C \operatorname{sgn}(\mathbf{v}_T) \quad (31)$$

which is, in fact, the Coulomb friction model. It must be highlighted that the Dahl model is not capable of capturing the Stribeck effect and stiction, since it is based on the dry Coulomb friction model with the introduction of pre-sliding displacement through a new state variable, which eliminates the discontinuity at zero velocity.

3.2 The bristle model

Haessig and Friedland [46] presented a model, which intends to simulate the randomness associated with frictional behavior of the contact between two surfaces with irregularities.

This model considers that the friction force is originated by the deformation of the asperities. Each contact is modeled as a bond between a rigid bristle and a massless bristle. The contact behaves as a spring, and when relative motion happens, the strain in the bristle increases as well as the friction force. The total friction force is given by the summation of each spring force,

$$\mathbf{F} = \sum_{i=1}^N \sigma_i (\mathbf{x}_i - \mathbf{b}_i) \quad (32)$$

where N denotes the number of bristles, σ_i represents the stiffness of the bristles, $\mathbf{x}_i$ is the relative position of the bristles and $\mathbf{b}_i$ is the location where the connection was formed. When the strain at a certain bristle exceeds its limit ($\|\mathbf{x}_i - \mathbf{b}_i\| > \delta_i$), the rupture of that bond occurs and a new bond is formed in a random location, which is calculated from the previous one.

The capacity of the bristle model to capture the random behavior highly depends on the number of bristles chosen. Instead of considering the real number of bonds, this approach only regards a much smaller number. The authors suggest less than 50. The higher the number of used bristles, the more complex the model becomes. This makes the model inefficient for computational simulations, and therefore, it is not usually employed.

3.3 The reset integrator model

The same authors of the bristle model [46] proposed also a more efficient and workable model, which is an evolution of the Dahl model. In this approach, the friction force is originated by the elastic and plastic deformations of the surface asperities. Each contact is modeled as a bond between two bristles. The reset integrator model does not allow for the bond to break, which means that when the strain of a connection increases until reaching the rupture point, the model ensures that it is kept constant. This model uses an extra state variable ($\mathbf{z}$) to determine the strain in the bond and to account the stiction, as

$$\frac{d\mathbf{z}}{dt} = \begin{cases} \mathbf{0} & \text{if } \|\mathbf{z}\| \geq z_0 \wedge \mathbf{z} \cdot \mathbf{v}_T > 0 \\ \mathbf{v}_T & \text{otherwise} \end{cases} \quad (33)$$

Similar to other friction models, the reset integrator model is also composed of two state equations, one for adhesion and another for sliding. The transition between those two phases occurs when the deflection reaches its maximum value z_0 . This friction force can then be defined as follows

$$\mathbf{F} = \begin{cases} \sigma_0(\mathbf{v}_T)(1+a)\mathbf{z} + \sigma_1 \frac{d\mathbf{z}}{dt} & \text{if } \|\mathbf{z}\| < z_0 \\ \sigma_0(\mathbf{v}_T)z_0 \text{sgn}(\mathbf{z}) & \text{if } \|\mathbf{z}\| \geq z_0 \end{cases} \quad (34)$$

where $\sigma_1 d\mathbf{z}/dt$ is the damping term that introduces some physical meaning by having damping oscillations and viscous friction effects, a denotes the coefficient

pertaining to the stiction and $\sigma_0(\mathbf{v}_T)$ is the contact stiffness. This friction force model has a discontinuity when the analysis changes between sticking and sliding situations.

3.4 Bliman–Sorine model

Bliman and Sorine [85–87] developed a family of dynamic friction models based on the approach proposed by Dahl [42]. In this model, the magnitude of friction force only depends on the sign of relative velocity and the space variable s , which produces a variable transformation,

$$ds = |v_T(t)| dt \quad (35)$$

This space variable can be defined as the absolute relative displacement of the contacting bodies since the last change in velocity. Hence, these models can be expressed as linear system,

$$\frac{dx_s}{ds} = Ax_s + Bv_s \quad (36)$$

$$F = Cx_s \quad (37)$$

where $v_s = \text{sgn}(v_T)$.

Bliman and Sorine [85–87] introduced a family of models which have variable complexity. For the first-order model, the constants are given by

$$A = -\frac{1}{\epsilon_f}, \quad B = \frac{f_1}{\epsilon_f} \quad \text{and} \quad C = 1 \quad (38)$$

Considering these values, the differential equation for the friction force can be written as

$$\frac{dF}{dt} = \frac{dF}{ds} \frac{ds}{dt} = \frac{dF}{ds} |v_T| = \frac{f_1}{\epsilon_f} \left(v_T - |v_T| \frac{F}{f_1} \right) \quad (39)$$

Thus, Eq. (39) is similar to Dahl model, if $F_C = f_1$, $\sigma = f_1/\epsilon_f$ and $\alpha = 1$. Bearing that in mind, this model is not capable of capturing stiction or the Stribeck effect. Although the second-order model allows including that phenomena, considering the following constants

$$A = \begin{pmatrix} -\frac{1}{\eta\epsilon_f} & 0 \\ 0 & -\frac{1}{\epsilon_f} \end{pmatrix}, \quad B = \begin{pmatrix} \frac{f_1}{\eta\epsilon_f} \\ -\frac{f_2}{\epsilon_f} \end{pmatrix} \quad \text{and} \quad C = (1 \ 1) \quad (40)$$

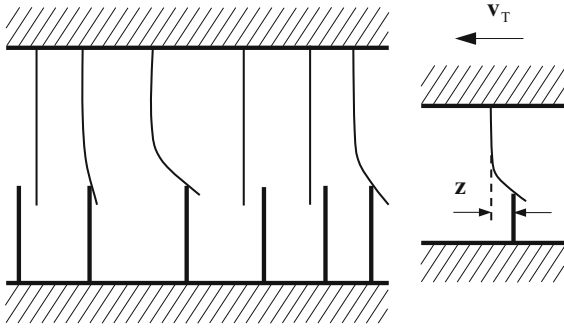


Fig. 12 Physical interpretation of LuGre model

This model works as a parallel connection of a fast and a slow Dahl model. The force generated by the slow model should be subtracted from the fast model.

3.5 LuGre model

The LuGre model was proposed by Canudas de Wit et al. [47] and can be considered as a derivation from the Dahl model [42]. This model is capable of capturing the Stribeck and stiction effects. In a simple way, this model considers friction as the result of the interactions of the surfaces bristles and utilizes the average bristle deflection. When a force is applied, the bristles start to deform with spring behavior during the sticking phase. Then if the force is sufficiently large, the bodies start to slip.

To quantify the average bristle deflection, it is necessary to introduce an internal state variable $\mathbf{z}$, as it is represented in Fig. 12. Thus, the model follows as

$$\frac{d\mathbf{z}}{dt} = \left(1 - \frac{\sigma_0}{g(\mathbf{v}_T)} \mathbf{z} \cdot \text{sgn}(\mathbf{v}_T)\right) \mathbf{v}_T \quad (41)$$

$$\mathbf{F} = \sigma_0 \mathbf{z} + \sigma_1(\mathbf{v}_T) \frac{d\mathbf{z}}{dt} + f(\mathbf{v}_T) \quad (42)$$

where σ_0 is the stiffness of the bristles, $\sigma_1(\mathbf{v}_T)$ is the damping of the bristles which can be set at constant, or can be a function of velocity, $f(\mathbf{v}_T)$ is an arbitrary function that describes the viscous effect and $g(\mathbf{v}_T)$ is an arbitrary function that accounts for the Stribeck effect as

$$g(\mathbf{v}_T) = F_C + (F_S - F_C) e^{-\left(\frac{\|\mathbf{v}_T\|}{v_S}\right)^2} \quad (43)$$

where F_C is the Coulomb friction, F_S is the static friction and v_S is the characteristic velocity of the Stribeck friction [35].

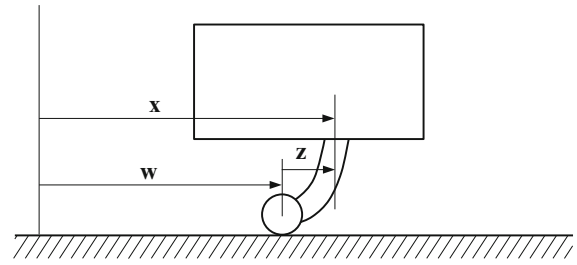


Fig. 13 Physical interpretation of the elastic and plastic displacements

For $f(\mathbf{v}_T)$, typically a linear viscous friction is considered, that is,

$$f(\mathbf{v}_T) = \sigma_2 \mathbf{v}_T \quad (44)$$

The effect of this term is quite important when there is a fluid lubricant or when the relative velocity is high.

For a constant velocity, that is, when the system reaches the steady state ($d\mathbf{z}/dt = \mathbf{0}$), the expression to the friction force can be reduced to

$$\mathbf{F} = g(\mathbf{v}_T) \text{sgn}(\mathbf{v}_T) + f(\mathbf{v}_T) \quad (45)$$

Thus, considering the Eqs. (43) and (44), it is possible to conclude that the steady-state values of the LuGre model agrees with the static friction model defined by Eq. (9).

3.6 Elasto-plastic model

The elasto-plastic model was developed by Dupont et al [48,88], and it is based on the division of the body's displacement $\mathbf{x}$ into two different components, the elastic and plastic displacements, $\mathbf{z}$ and $\mathbf{w}$. When the bodies are sticking, the plastic displacement remains constant, while during the sliding phase, the elastic displacement is constant. Figure 13 shows the physical representation of each of these two types of displacement, and the elastic displacement represents the bristle deflection.

The friction force can be calculated in a similar way as compared to the case of LuGre model [47],

$$\mathbf{F} = \sigma_0 \mathbf{z} + \sigma_1 \frac{d\mathbf{z}}{dt} + \sigma_2 \mathbf{v}_T \quad (46)$$

where σ_0 is the contact stiffness, σ_1 denotes the contact damping and σ_2 represents the viscosity coefficient. The velocity of bristle deflection is given by

$$\frac{dz}{dt} = v_T \left(1 - \alpha(z, v_T) \frac{\sigma_0}{g(v_T)} z \cdot \text{sgn}(v_T) \right) \quad (47)$$

in which the function $\alpha(z, v_T)$ is used to capture stiction, since it just allows elastic displacement until the system reaches the breakaway force. This function is described as

$$\alpha(z, v_T) = \begin{cases} \alpha(z) & \text{if } v_T \cdot z \geq 0 \\ 0 & \text{if } v_T \cdot z < 0 \end{cases} \quad (48)$$

with

$$\alpha(z) = \begin{cases} 0 & \text{if } \|z\| < z_{ba} \\ \frac{1}{2} \left(\sin \left(\pi \frac{\|z\| - z_{ba}}{z_{max} - z_{ba}} \right) + 1 \right) & \text{if } z_{ba} < \|z\| < z_{max} \\ 1 & \text{if } z_{max} < \|z\| \end{cases} \quad (49)$$

where z_{max} is the maximum bristle deflection and z_{ba} is the breakaway bristle deflection. The value of z_{max} can be determined based on the steady-state friction, and the relation $z_{ba}/z_{max} \approx 0.7$ can be used to calculate z_{ba} [48]. The function $g(v)$ describes the Stribeck curve and similarly to LuGre is given by Eq. (43).

3.7 Leuven model

This model was proposed by Swevers et al. [49] as being an improvement over the LuGre model [47]. The Leuven model can capture the Stribeck effect, frictional lag, breakaway force variation, stick-slip behavior and pre-sliding hysteresis with nonlocal memory. The latter property is not modeled by the LuGre model.

Leuven model also considers the average deflection of the asperities, z , and it can be defined by

$$\frac{dz}{dt} = v_T \left(1 - \text{sgn} \left(\frac{F_d(z)}{g(v_T) - F_b} \right) \left| \frac{F_d(z)}{g(v_T) - F_b} \right|^n \right) \quad (50)$$

$$F = F_h(z) + \sigma_1 \frac{dz}{dt} + \sigma_2 v_T \quad (51)$$

where n represents the coefficient that defines the shape of the transition curve and $g(v_T)$ is an arbitrary function that models the steady-state behavior of friction force, which is commonly described in a similar way to Eq. (43) by

$$g(v_T) = \left(F_C + (F_S - F_C) e^{-\left| \frac{v_T}{v_S} \right|^\delta} \right) \text{sgn}(v_T) \quad (52)$$

The hysteresis friction force is represented by $F_h(z)$, and it is constituted by two components as follows

$$F_h(z) = F_b + F_d(z) \quad (53)$$

where F_b is the hysteresis friction force at the beginning of a transition curve when there is a velocity reversal and $F_d(z)$ represents the transition curve active at a certain time.

The Leuven model presents a good characterization of friction phenomena; nevertheless, it has also several implementation problems and a discontinuity in the friction force for some circumstances. Few years later, the same authors [89] proposed some modifications to the model. The discontinuity issue was overcome by changing the state equation to the following form

$$\frac{dz}{dt} = v_T \left(1 - \text{sgn} \left(\frac{F_h(z)}{g(v_T)} \right) \left| \frac{F_h(z)}{g(v_T)} \right|^n \right) \quad (54)$$

3.8 Generalized Maxwell slip model

The generalized Maxwell slip model [50, 90] is a development of Leuven model [49], and it considers a parallel connection of N single-state friction models, as shown in Fig. 14. They are described by the same dynamical model and have the same input displacement and velocity, but with different sets of parameter values. Each single-state friction model is evaluated separately to check whether it is sticking or sliding.

Thus, if the element i is sticking, it is governed by the following equation

$$\frac{dz_i}{dt} = v_T \quad (55)$$

which happens until $z_i = g_i(v_T)$, then for sliding, the state equation is written as

$$\frac{dz_i}{dt} = \text{sgn}(v_T) C_i \left(1 - \frac{z_i}{g_i(v_T)} \right) \quad (56)$$

and it is valid until the relative velocity passes through zero. From the previous equations, z_i denotes the spring

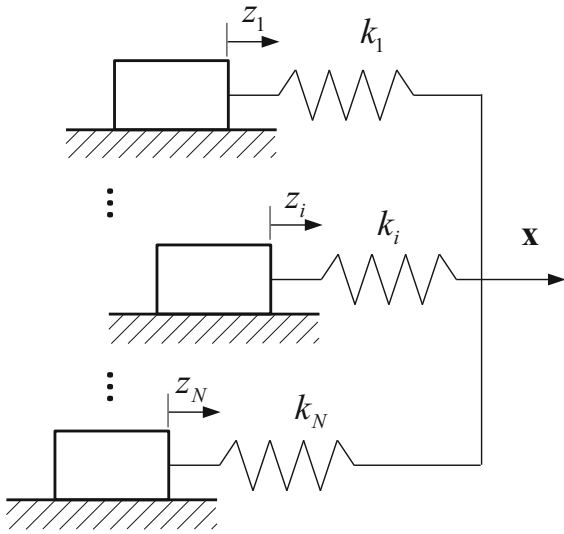


Fig. 14 Representation of the generalized Maxwell slip model

deflection, $g_i(v_T)$ is the Stribeck curve for the deformation of element i and C_i is the attraction parameter which determines how fast z_i converges to $g_i(v_T)$.

To evaluate the total friction force, the contribution of each single-state model should be summed, plus the viscous friction component, as it follows

$$F_f(t) = \sum_{i=1}^N \left(k_i z_i + \sigma_i \frac{dz_i}{dt} \right) + f(v_T) \quad (57)$$

where k_i is the spring stiffness, σ_i is damping coefficient and $f(v_T)$ is the viscous component which is normally set with a linear dependence of velocity as in Eq. (44).

Hence, to implement this model, it is necessary to define three parameters per element besides the Stribeck function, which normally uses three additional parameters. However, a scale parameter, α_i , can be introduced which can be used to calculate the attraction parameter for each element based on a global value, i.e., $C_i = \alpha_i C$. The scaling parameters can also be applied for the Stribeck functions and the damping coefficients. Though, these parameters should guarantee the following relation

$$\sum_{i=1}^N \alpha_i = 1 \quad (58)$$

Thus, the model is reduced to two parameters per element, and six for the whole system. It was shown by Piatkowski [91] that using only four elements could provide good results.

3.9 Gonthier et al. model

Gonthier et al. [52] introduced a two-dimensional friction model based on LuGre approach [47]. The authors considered a force from the bending of the bristles, given by

$$\mathbf{F}_{br} = \sigma_0 \mathbf{z} + \sigma_1 \frac{d\mathbf{z}}{dt} \quad (59)$$

where σ_0 is the stiffness and σ_1 is the damping coefficient. To ensure a smooth transition between the stick–slip friction regimes, an auxiliary parameter is defined as,

$$s = e^{-\left(\frac{\|\mathbf{v}_T\|}{v_S}\right)^2} \quad (60)$$

where v_S is the Stribeck velocity. When the bodies are sticking, the deformation rate will be equal to the relative velocity, while for sliding, the resultant friction force will approach the Coulomb friction force, $\mathbf{F}_C$, as follows

$$\frac{d\mathbf{z}}{dt} = s \mathbf{v}_T + (1 - s) \left(\frac{1}{\sigma_1} \mathbf{F}_C - \frac{\sigma_0}{\sigma_1} \mathbf{z} \right) \quad (61)$$

Coulomb friction always possesses the velocity direction and can be approximated by

$$\mathbf{F}_C = F_C \text{dir}_\epsilon(\mathbf{v}_T, v_\epsilon) \quad (62)$$

where $\text{dir}_\epsilon(\mathbf{v}_T, v_\epsilon)$ returns the unit vector with velocity direction, and it smooths the vector oscillations for velocities under a certain limit, v_ϵ , to diminish the discontinuities in velocity direction. This tolerance velocity is considered $v_\epsilon = 0.01 v_S$.

$$\text{dir}_\epsilon(\mathbf{v}_T, v_\epsilon) = \begin{cases} \frac{\mathbf{v}_T}{\|\mathbf{v}_T\|} & \text{if } \|\mathbf{v}_T\| \geq v_\epsilon \\ \frac{\mathbf{v}_T}{v_\epsilon} \left(\frac{3}{2} \frac{\|\mathbf{v}_T\|}{v_\epsilon} - \frac{1}{2} \left(\frac{\|\mathbf{v}_T\|}{v_\epsilon} \right)^3 \right) & \text{if } \|\mathbf{v}_T\| < v_\epsilon \end{cases} \quad (63)$$

This approach includes a temporal lag associated with the dwell-time dependence. To capture that phenomenon, a new state variable is defined as

$$\dot{s}_{dw} = \begin{cases} \frac{1}{\tau_{dw}} (s - s_{dw}) & \text{if } s - s_{dw} \geq 0 \\ \frac{1}{\tau_{br}} (s - s_{dw}) & \text{if } s - s_{dw} < 0 \end{cases} \quad (64)$$

where τ_{dw} is the dwell-time dynamics time constant and $\tau_{br} = \sigma_1/\sigma_0$ is the bristle dynamics time constant. The time constants should be set according to the desired time delay, a large one for sticking and a small time delay for sliding. Thus, the maximum friction force can be defined as

$$F_{\max} = F_C + (F_S - F_C) s_{dw} \quad (65)$$

where F_C and F_S are the magnitude of Coulomb and static friction, respectively. Thus, the friction force can be expressed as

$$\mathbf{F} = \begin{cases} \mathbf{F}_{br} + \sigma_2 \mathbf{v}_T & \text{if } \|\mathbf{F}_{br}\| \leq F_{\max} \\ F_{\max} \text{sgn}(\mathbf{F}_{br}) + \sigma_2 \mathbf{v}_T & \text{if } \|\mathbf{F}_{br}\| > F_{\max} \end{cases} \quad (66)$$

where σ_2 is the viscous damping coefficient. The use of this model results in a set of ordinary differential equations which are quite stiff at low relative velocities but which cannot be solved using explicit ODE solvers.

3.10 Liang et al. bristle model

This bristle friction model [55] is an extension of the model presented by Heissig and Friedland [46] to the 3D space. The average deflection of the bristles is represented by a linear spring, which can stretch and rotate, and it is constrained to the tangential plane of the contact.

An illustration of the tangential plane for a single contact is represented in Fig. 15. The central point P represents the contact point, where the normal and friction forces are applied. Thus, for each individual contact, the friction force can be calculated as

$$\mathbf{F} = k_b \mathbf{z} \quad (67)$$

where k_b is the bristle stiffness and $\mathbf{z}$ is the average bristle deflection that can be expressed as

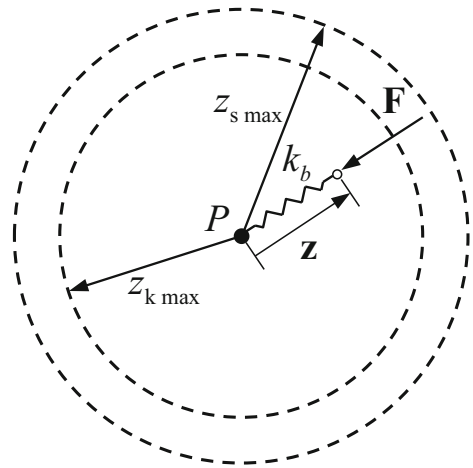


Fig. 15 Representation of the Liang et al. bristle model

$$\mathbf{z}(t) = \begin{cases} \mathbf{z}(t_0) + \int_{t_0}^t \mathbf{v}_T(t) dt & \text{if } \|\mathbf{z}\| < z_{\max} \\ z_{\max} \text{sgn}(\mathbf{v}_T) & \text{if } \|\mathbf{z}\| \geq z_{\max} \end{cases} \quad (68)$$

where t_0 is the starting time of the contact, t is the current time and $z_{\max}$ is the maximum bristle deflection that can be defined as

$$z_{\max} = \begin{cases} z_{k \max} = \frac{F_C}{k_b} & \text{if } \|\mathbf{v}_T\| > v_d \\ z_{s \max} = \frac{F_S}{k_b} & \text{if } \|\mathbf{v}_T\| \leq v_d \end{cases} \quad (69)$$

where v_d is a threshold velocity, which represents the numerical boundary between the adhesion and sliding regimes. Thus, there is a maximum value for bristle deflection for adhesion ($z_{s \max}$) and another for sliding ($z_{k \max}$), and both parameters are represented in Fig. 15.

The major drawback of this model is related to the transition of the sticking regime to the sliding regime, due to a sudden decrease in the maximum value of the average bristle deflection, which can result in an abrupt change in the friction force. To stabilize the friction force, Eq. (67) can be modified through the introduction of a damping term, i.e.,

$$\mathbf{F} = k_b \mathbf{z} + c_b \dot{\mathbf{z}} \quad (70)$$

where c_b is the bristle damping coefficient. When the model reaches the steady state in the sliding mode, the friction force will be equal to the Coulomb friction force. In the sticking mode, the friction force will be higher and equal to the static friction.

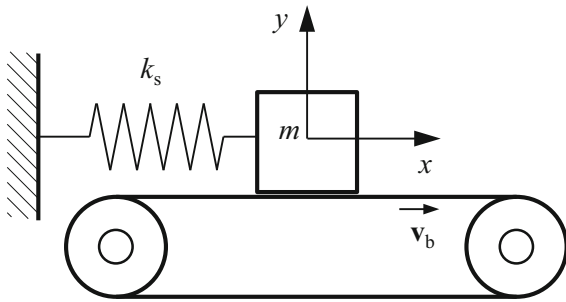


Fig. 16 Representation of the 1-DoF mass–spring model on a conveyor belt with sliding friction

4 Comparison of friction force models using demonstrative application examples

The goal of this section is to examine the influence of using different friction force models on the dynamics response of multibody systems. For this, two application examples are utilized, namely a simple mass–spring model with one degree of freedom and a multi-body system with a frictional translational joint.

4.1 Single DOF mass–spring on a conveyor system with sliding friction

The classic sliding mass–spring system is first utilized as a numerical application example for the analysis and comparison of some of the previously presented friction models (see Fig. 16). This system is widely used as benchmark for the validation of friction models [6, 19, 20, 29, 46, 52, 92, 93], and it consists of a block with mass m , which is positioned on a conveyor belt. The belt is moving with a constant velocity. The block is connected by a spring element with stiffness k_s .

The simulation parameters for the spring-mass system are presented in Table 1. The specific param-

eters associated with each friction model were extracted from the literature and are listed in Table 2. Initially, the block is located at the origin of the xy coordinate system, and its velocity being the same as the belt.

Figures 17, 18 and 19 show the main results obtained from the simulations with different friction models. In order to keep the analysis simple, the frictional approaches are grouped into three classes, namely “static models without stiction,” “static models with stiction” and “dynamic models.” The behavior of the system is quantified by the plots of position, relative velocity and friction force values. The results correspond to 20 s of numerical simulation.

The static models without stiction present an adhesion phase related to the initial conditions, since the spring force magnitude is lower than the Coulomb friction force. The fact of having an exact zero relative velocity at the beginning of the simulation avoids any numerical instability for the models with a discontinuity at null velocity, as in the case of the Coulomb model and the Coulomb with viscous approach. Since these models have a constant Coulomb friction force or lower for velocities close to zero, the block exhibits a spring-like behavior with a frequency equal to $1/2\pi\sqrt{k_s/m}$. From the analysis of Fig. 17, it can be observed that the results corresponding to Coulomb with viscous friction show a distinct behavior since the viscosity introduces a damping effect in the block oscillations.

For the static models with stiction, the differences are more evident. Both simulations with Coulomb with stiction and Stribeck friction present numerical instability, which can be observed in the friction force plot of Fig. 18. This occurs due to the changes in the velocity direction. The Coulomb model with stiction is capable of reaching the static friction only at the first peak, and because of the initial velocity, it does not stick again. Karnopp and Awrejcewicz models have similar responses and present well-defined stick and slip phases. As the model with Stribeck friction is the only one with viscous friction component, it has faster cycles comparing with the remaining approaches.

Analyzing the dynamic models, it is possible to observe that only Dahl’s model does not capture the stick–slip phenomenon. From Fig. 19, the remaining models can be divided into two groups, the first includes LuGre, elasto-plastic and Gonthier and the second one includes Reset Integrator and Liang. The difference in the dynamic behavior of these models is caused by the

Table 1 Simulation parameters for the spring-mass model

Parameter	Symbol	Value	Parameter	Value
Mass of the block	m	1 kg	Time step	0.00005 s
Velocity of the belt	v_b	0.1 m/s	Simulation time	20 s
Spring stiffness	k_s	2 N/m	Integrator scheme	Runge–Kutta fourth order

Table 2 Parameters considered for the different friction models

Parameter	S	Value	Parameter	S	Value
Static coefficient of friction	μ_s	0.15	Factor for curve shape	ξ	50 s/m
Kinetic coefficient of friction	μ_k	0.1	Stiction coefficient	a	0.5
Coefficient of viscosity	F_v	0.1 Ns/m	Stiffness coefficient	σ_0	10^5 N/m
Stribeck velocity	v_s	0.001 m/s	Damping coefficient	σ_1	$\sqrt{10^5}$ Ns/m
Geometry factor	δ_σ	2	Coefficient of viscosity	σ_2	0.1 Ns/m
Tolerance velocity (Karnopp)	D_v	0.001 m/s	Dwell-time constant	τ_{dw}	2 s
Tolerance velocity (Threlfall, Bengisu and Akay, Awrejcewicz)	v_0	0.001 m/s	Bristle stiffness	k_b	50000 N/m
Tolerance velocity (Ambrósio)	v_0	0.0001 m/s	Threshold velocity	v_d	0.001 m/s
Tolerance velocity (Ambrósio)	v_1	0.001 m/s	Bristle damping	c_b	80 Ns/m

presence of a viscous component in the first set of models.

Figures 20 and 21 show the plots of the friction force versus displacement and friction force versus relative velocity for the dynamic friction approaches. It should be noted that the displacement displayed in Fig. 20 is not the same as the position plotted in Figs. 17, 18 and 19, since the position represents the location of the block in the global reference frame, while the displacement characterizes the relative displacement of the contacting surfaces from the beginning of the simulation. Comparing these models with Rabinowicz's work [4], it can be stated that LuGre, elasto-plastic and Gonthier's present the friction force as a function of displacement with a similar shape. During the pre-sliding displacement phase, the slope of the friction force is directly related to the stiffness coefficient of each model (see Fig. 20). In contrast to the static models, the dynamic approaches do not change the force direction with the change of velocity direction, as can be observed in Fig. 21. This behavior is similar to the one obtained in [93]. The Gonthier, elasto-plastic and LuGre (to some extent) seem to increase the friction force before sticking. In turn, the Liang's model shows more instability in velocity, before reaching the sticking phase. From the analysis of Fig. 21, it is also possible to state

that LuGre and elasto-plastic can describe the frictional lag phenomenon, since both models present a higher friction force when the relative velocity is increasing.

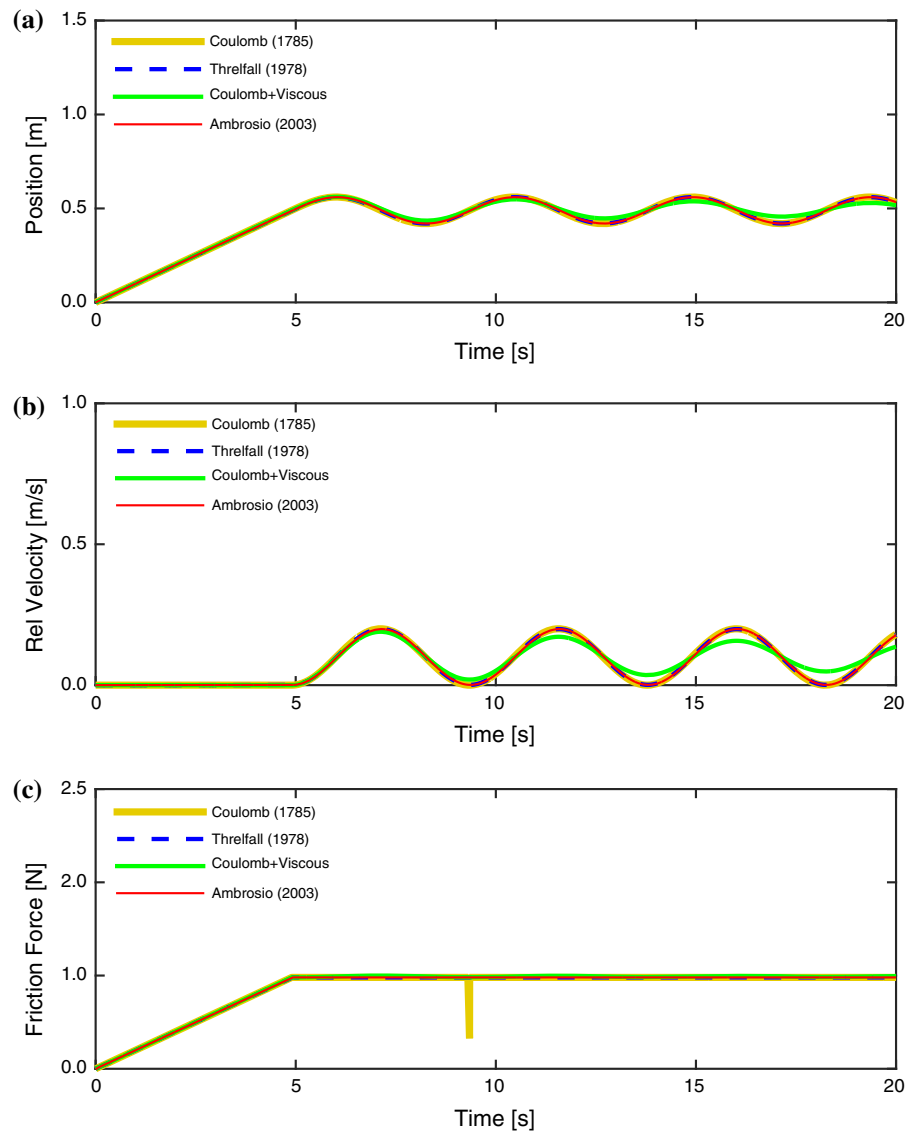
4.1.1 Computational efficiency of friction force models for the sliding 1D mass-spring system

In order to present a more comprehensive comparison of the friction force models, the study has been extended to include the computational efficiency and the effect of some of frictional parameters.

For the evaluation of efficiency of each friction model, the computing time has been measured using different MATLAB's integration algorithms, namely *ode45* and *ode15s*. The former is one of the most general solvers for differential equations, and the latter is more suitable for stiff problems. The simulations were performed with a reporting time step of 5×10^{-5} s, and a relative error tolerance of 10^{-8} was specified. The analysis of the efficiency is conducted by comparing the normalized computing time, in which the fastest simulation is used as reference, and calculated as

$$\psi_i = \frac{T_i}{T_{\min}} \quad (71)$$

Fig. 17 Results from simulations with static models without stiction



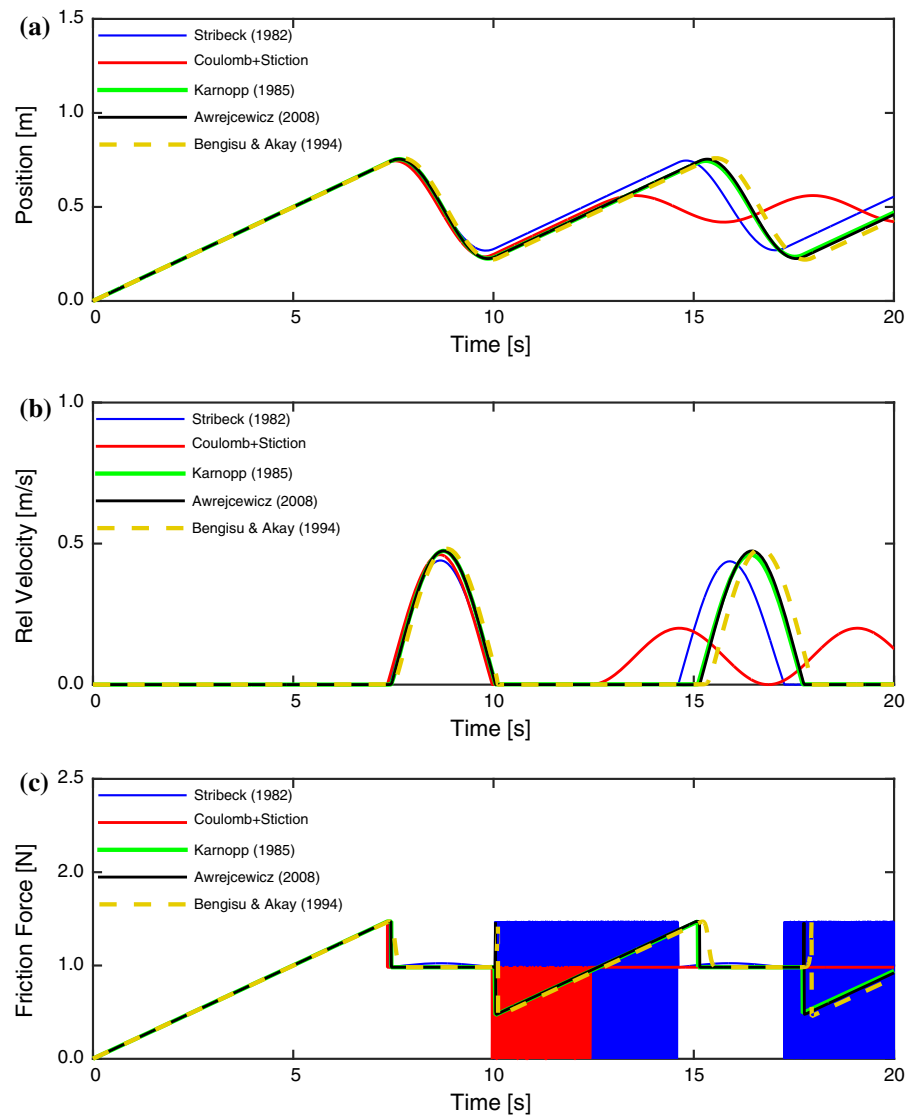
where T_i is the computing time for the model to evaluate and $T_{\min}$ denotes the computing time of the fastest simulation. Table 3 shows the results for the computational efficiency. One can notice that two friction force models were not able to successfully simulate the 1-DoF mass–spring model with a variable time step due to the numerical discontinuity that they present for zero velocity. From the remaining models and results, it is clear that the utilization of *ode15s* significantly diminishes the computing time, which means that the modeling of frictional effects increases the stiffness of the problem to be solved. Generally, the dynamic friction models require more computational effort than the sta-

tic ones. In particular, the Dahl's model is the most time-consuming model, especially with the stiff solver. Overall the results indicate that the computational efficiency performance agrees with the more comprehensive study undertaken by Pennestrì et al. [75], in which more solvers and different tolerances were tested.

4.1.2 Parametric study of friction force models for the sliding 1D mass–spring system

The choice of a friction model to implement in a dynamic simulation is not an easy task. Nevertheless, in order to have more complex friction models, it is in gen-

Fig. 18 Results from simulations with static models with stiction

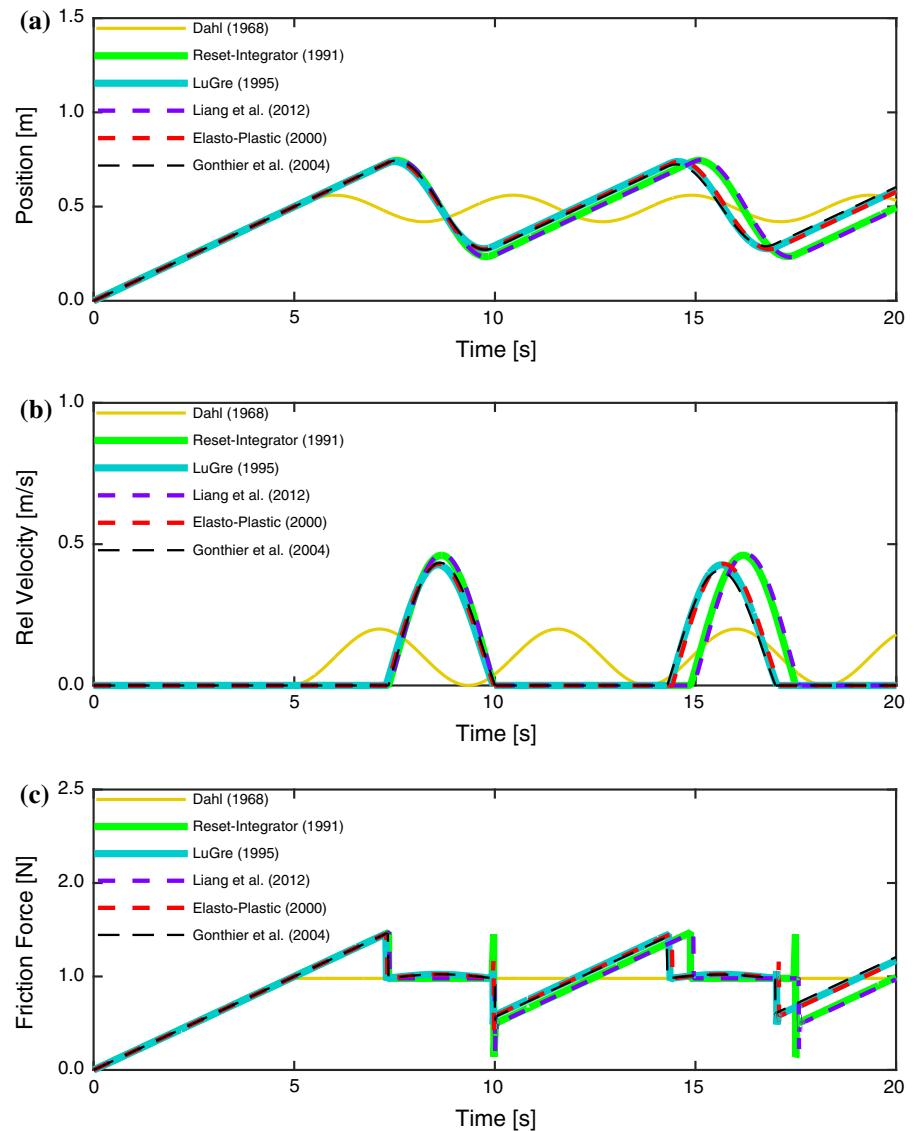


eral necessary to introduce larger number of parameters to fully describe the physics of the friction phenomena. Sometimes, these parameters are obtained experimentally, which could be a significant limitation for modeling approaches that need the determination of many parameters. However, some of them are chosen empirically, since they cannot be physically measured. Some studies have been conducted to examine the friction modeling through the analysis and comparison of different approaches [30,32,74,75], although only few of them have been paying attention to the influence of the parameters variation [93,94]. Most of utilized parameters are shared by several friction force models, but

they can result in different behaviors depending on the considered approach.

In the following, an analysis of the influence of several parameters of the most relevant dynamic friction force models utilized in multibody dynamics is presented. For this, the Dahl, Reset Integrator, LuGre and Gonthier models are investigated. The static models are left out of this study, since they are usually defined by a few number of parameters. Both the dynamic response of the 1-DoF mass-spring model and the computational efficiency are assessed for different input parameters. The results of the dynamic simulations are analyzed through the plots of the friction force versus dis-

Fig. 19 Results from simulations with dynamic models



placement and friction force versus relative velocity, as depicted in Figs. 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, and 34. These plots allow the identification of some frictional phenomena, such as the pre-sliding displacement or frictional lag, which cannot be easily recognized by the kinematic outputs as the position or velocity of the block. For the sake of simplicity, the coefficients of friction, either kinetic or static, are not in the scope of this study, since their influence in the friction force has been examined extensively in the literature. Each parameter has been set to half and twice the original value in order to examine how they affect the output motion of the system.

As described previously, the Dahl model utilizes three parameters to be fully defined. Since one of them is the kinetic coefficient of friction, only the curve shape parameter and the stiffness coefficient will be addressed. The analysis of the curve shape parameter based on the materials in contact is represented in Fig. 22a, b. A small value of α corresponds to a more brittle material behavior, while for the larger one, a more ductile behavior is expected. This can be corroborated by Fig. 22a, which shows a slower transition between the elastic and plastic phases of the contact, when this parameter is increased. Figure 23a, b shows the influence of the stiffness coefficient σ_0 which is

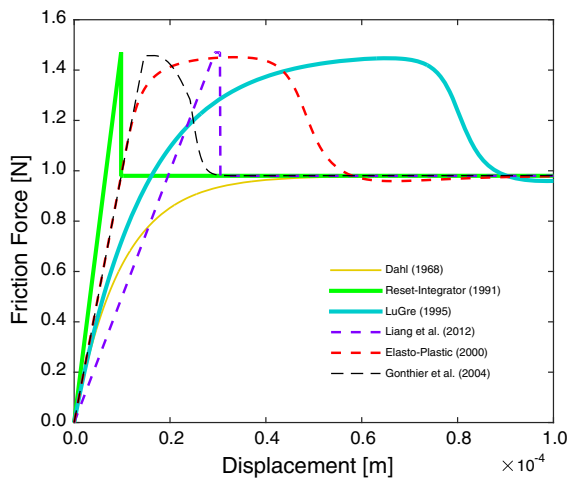


Fig. 20 Behavior of the dynamic models: friction force versus displacement

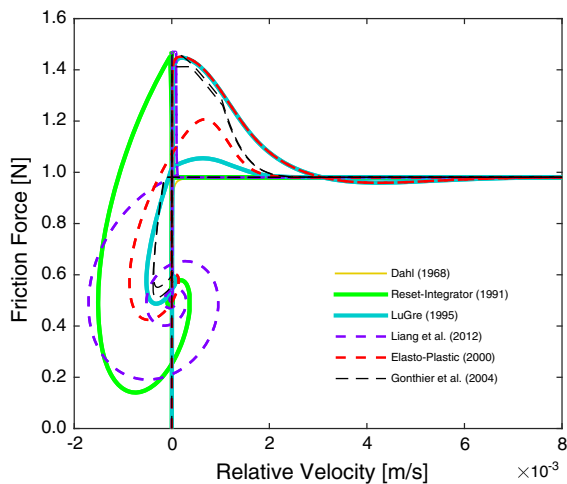


Fig. 21 Behavior of the dynamic models: friction force versus relative velocity

associated with the slope of the pre-sliding displacement. Moreover, the increase in this coefficient diminishes the velocity oscillations in the sticking phase, as observed from Fig. 23b.

The stiffness and damping coefficients (σ_0 and σ_1) are two of the four parameters utilized by the reset integrator model, and their influence is shown in Figs. 24 and 25, respectively. Similar to the previous model, the stiffness coefficient affects the slope of the pre-sliding displacement curve. In the friction–velocity relation, with the increase in the stiffness, the oscillation of the velocity decreases, but the variation of friction force is higher. In what concerns the damping coeffi-

Table 3 Normalized computing time ψ for the 1-DoF mass–spring model

Model	<i>ode45</i>	<i>ode15s</i>
Coulomb	1.79	1.03
Coulomb + Viscous	1.68	1.00
Coulomb + Stiction	*	*
Stribeck	*	*
Karnopp	2.59	1.33
Threlfall	26.86	1.22
Bengisu & Akay	88.09	2.31
Ambrósio	18.77	1.21
Awrejcewicz et al.	2.73	1.80
Dahl	490.37	14.47
Reset Integrator	24.94	2.35
LuGre	474.42	3.47
Elasto-plastic	482.32	3.19
Gonthier et al.	24.89	3.61
Liang et al.	30.59	2.74

* The model was not able to converge with the MATLAB's integration algorithm

cient, it does not affect the friction–displacement relation, although the reduction in this parameter largely increases the size of the sticking oscillation cycle, as shown in Fig. 25b.

As presented previously, the LuGre model has six different parameters. The stiffness, damping and viscosity coefficients, as well as the Stribeck velocity, will be analyzed. Figure 26a shows that the stiffness coefficient σ_0 producing the same result as for the Dahl and reset integrator models, by controlling the slope of the pre-sliding displacement, and therefore increasing the displacement range of the sticking phase. The increase in this parameter diminishes the effect of frictional lag by having higher friction force when the velocity approaches zero, as shown in Fig. 26b. The damping coefficient σ_1 does not change the pre-sliding displacement, although the lower the parameter, the smoother the transition from sticking to sliding regime, as depicted in Fig. 27a. With a higher damping coefficient, the friction force can even reach a value lower than the Coulomb friction, before starting the sliding phase. Similarly for the reset integrator model, as observed in Fig. 27b, this parameter has a direct influence on the size of the friction–velocity oscillation in the sticking mode.

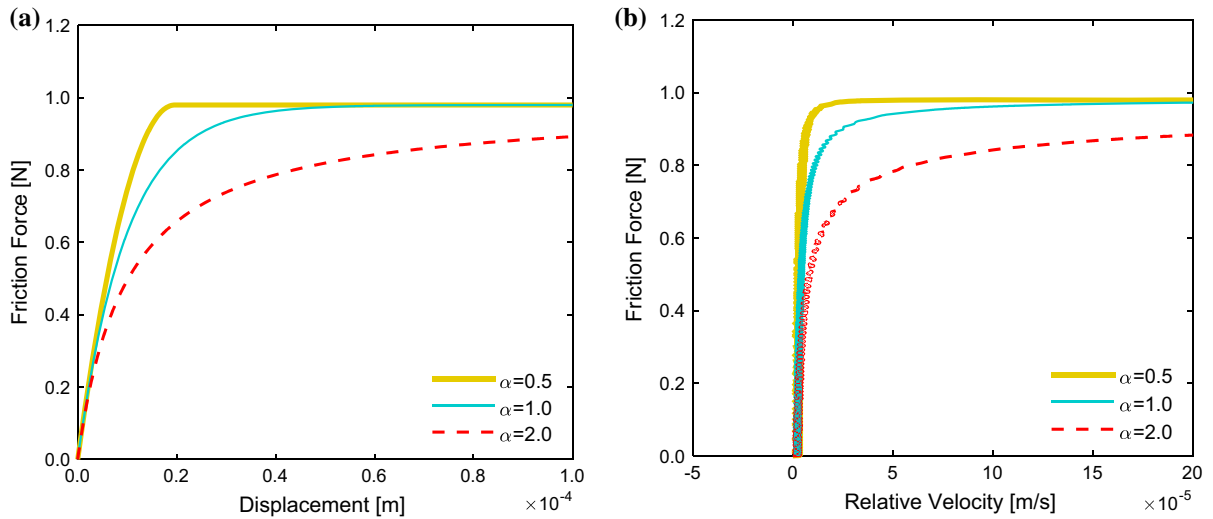


Fig. 22 Influence of curve shape parameter (α) in Dahl model: **a** friction force versus displacement; **b** friction force versus relative velocity

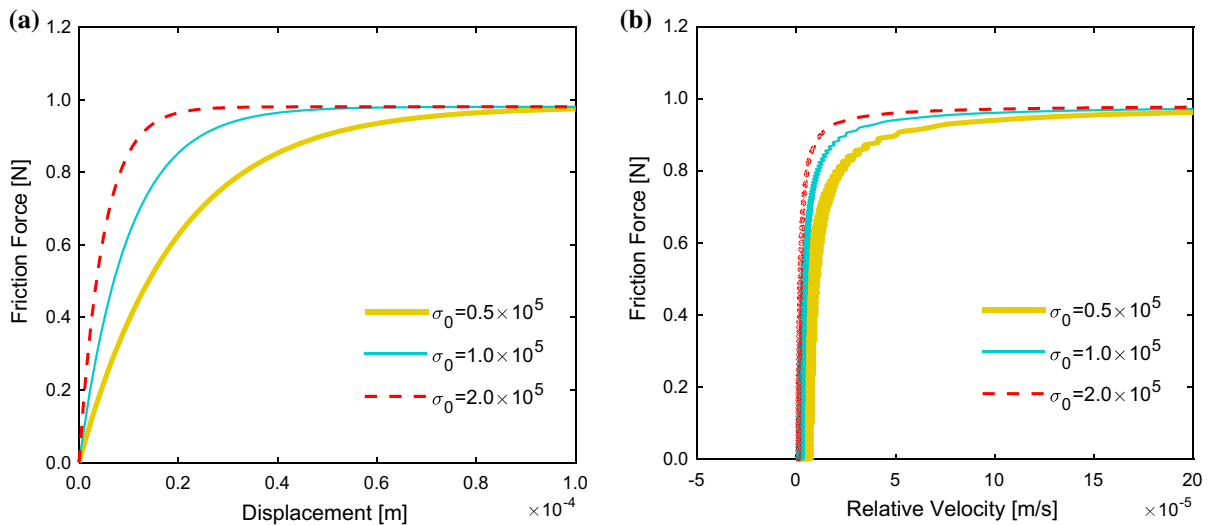


Fig. 23 Influence of stiffness coefficient (σ_0) in Dahl model: **a** friction force versus displacement; **b** friction force versus relative velocity

The viscosity coefficient σ_2 does not influence the friction–displacement relation, but it mitigates the friction force oscillation in the sticking mode, as represented in Fig. 28b. As it was expected, the increase in the Stribeck velocity increases the dimension of the sticking phase. Figure 29b shows that, for a high Stribeck velocity v_S , it is easier to detect the growth on the friction force when the velocity approaches zero.

The Gonthier model utilizes seven parameters to be fully implemented. The influence of five of them is assessed through the plots of Figs. 30, 31, 32, 33 and 34. The effects of stiffness coefficient σ_0 corroborate the analogous parameter in the previous models, as it affects the slope of the pre-sliding displacement. In contrast to LuGre model, the increase in the damping coefficient σ_1 makes the transition from sticking to sliding smoother, as shown in Fig. 31a.

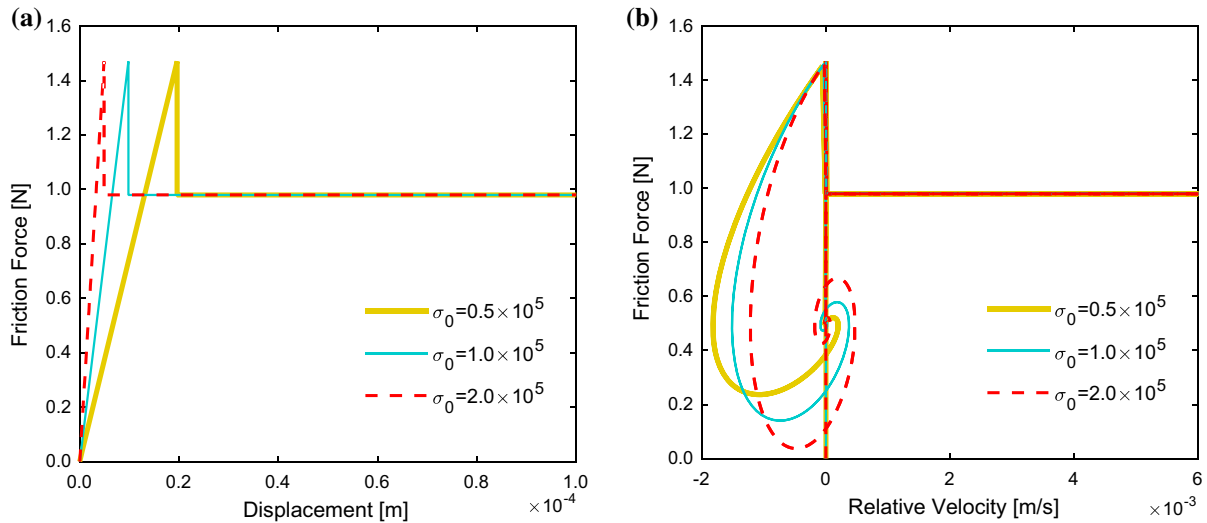


Fig. 24 Influence of stiffness coefficient (σ_0) in Reset Integrator model: **a** friction force versus displacement; **b** friction force versus relative velocity

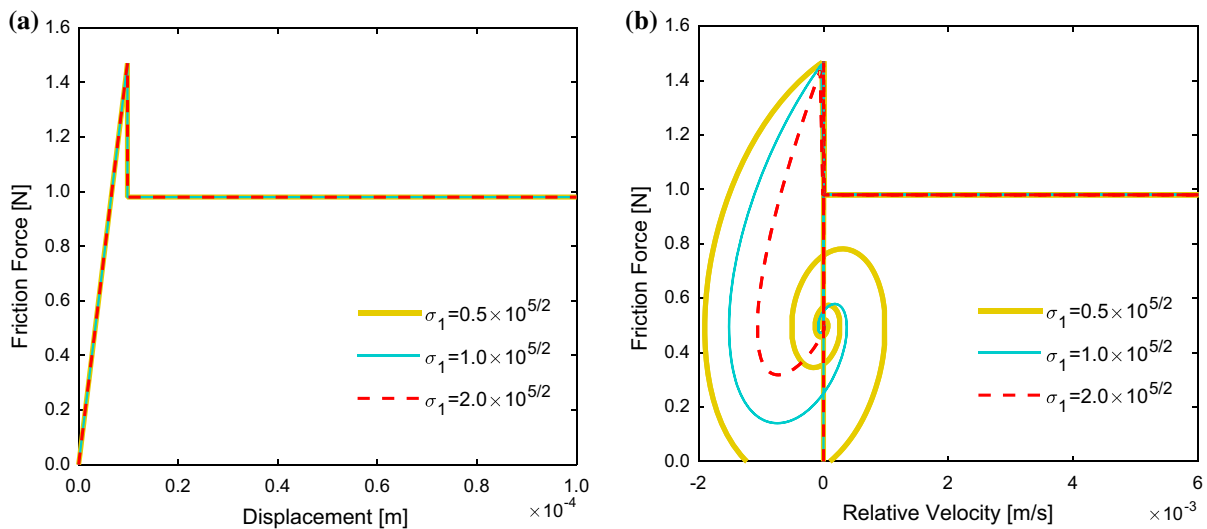


Fig. 25 Influence of damping coefficient (σ_1) in Reset Integrator model: **a** friction force versus displacement; **b** friction force versus relative velocity

Moreover, as in LuGre model, the viscosity coefficient σ_2 does not affect the friction–displacement relation, but it reduces the oscillation of the friction force during the sticking phase. The Stribeck velocity v_s enlarges the duration of the sticking phase, as depicted in Fig. 33. This parameter also increases the oscillation of the relative velocity during the sticking regime. Finally, the dwell-time constant τ_{dw} affects the maximum value of friction force, as the increase in this

parameter diminishes the maximum friction force, but it enlarges the duration of sticking phase, as illustrated in Fig. 34.

In addition to the analysis of the influence of some parameters in the modeling of the friction force, their impact on the computational efficiency was assessed as well. The computing time of each simulation has been measured, and a normalized variation of the computing time has been calculated based on the difference

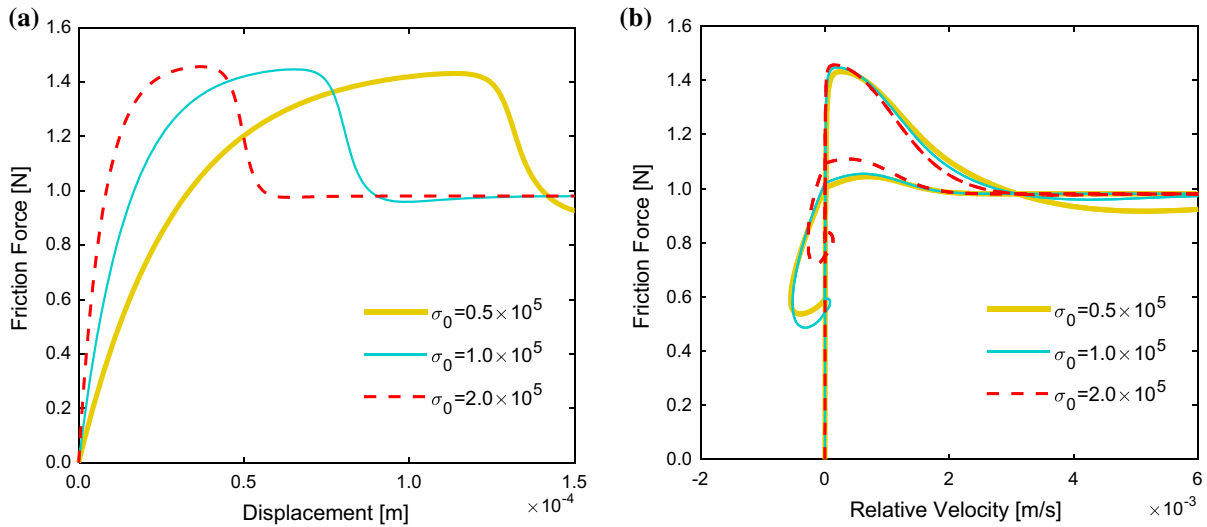


Fig. 26 Influence of stiffness coefficient (σ_0) in LuGre model: **a** friction force versus displacement; **b** friction force versus relative velocity

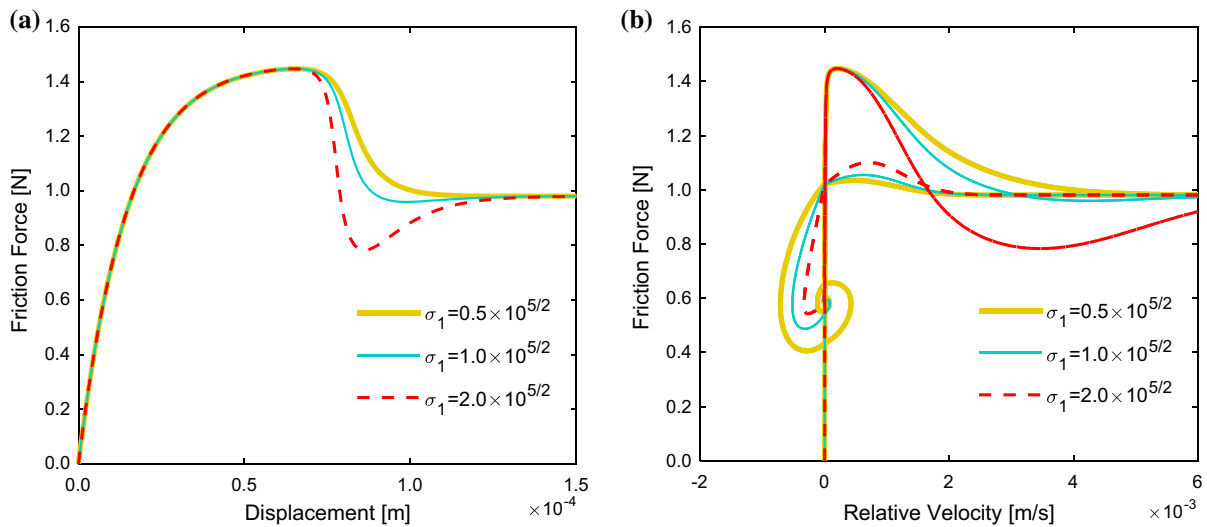


Fig. 27 Influence of damping coefficient (σ_1) in LuGre model: **a** friction force versus displacement; **b** friction force versus relative velocity

between the simulations with half and twice of the reference value, as follows

$$d\psi_i = \frac{T_{i,2} - T_{i,0.5}}{T_i} \quad (72)$$

in which $T_{i,0.5}$ and $T_{i,2}$ represent the computational time for the half and the twice the reference value, respectively. In that sense, having high absolute values indicates that the parameter has high impact

in the efficiency; on the other hand, values close to zero suggest that the variation is not significant.

Table 4 presents the normalized variation of the computing time for the previously discussed models, and their respective parameters. The parameter which has the most impact on computational efficiency is distinctly the curve shape parameter used by the Dahl's model, which requires more effort for low values in order to describe the behavior of

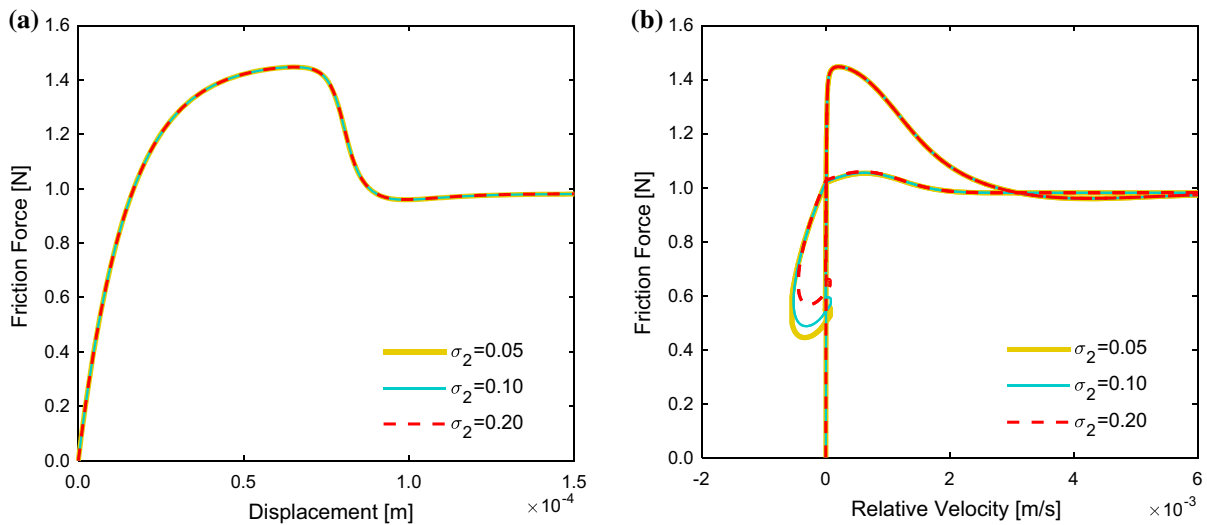


Fig. 28 Influence of viscosity coefficient (σ_2) in LuGre model: **a** friction force versus displacement; **b** friction force versus relative velocity

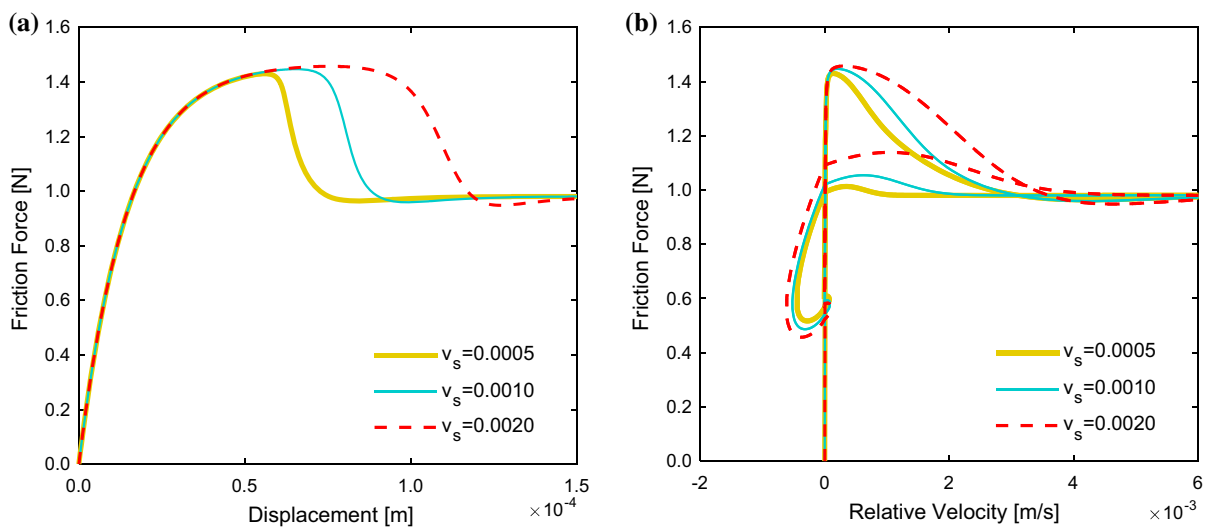


Fig. 29 Influence of Stribeck velocity (v_s) in LuGre model: **a** friction force versus displacement; **b** friction force versus relative velocity

brittle materials. One can state that high values for the stiffness coefficient demand more computational load, although this effect is mitigated by the utilization of a solver for stiff cases. Contrarily, the increase in the damping coefficient decreases the time of calculation, and in most cases, the *ode15s* does not enlarge the efficiency. The remaining parameters do not substantially affect the computational time.

4.2 Spatial slider-crank mechanism with frictional translational joint

In this section, a study is conducted on the influence of friction modeling in a kinematic translational joint for a spatial multibody system. For this application example, no clearance is considered in the sliding motion and the generated forces result from continuous contact. Before presenting this application example, the

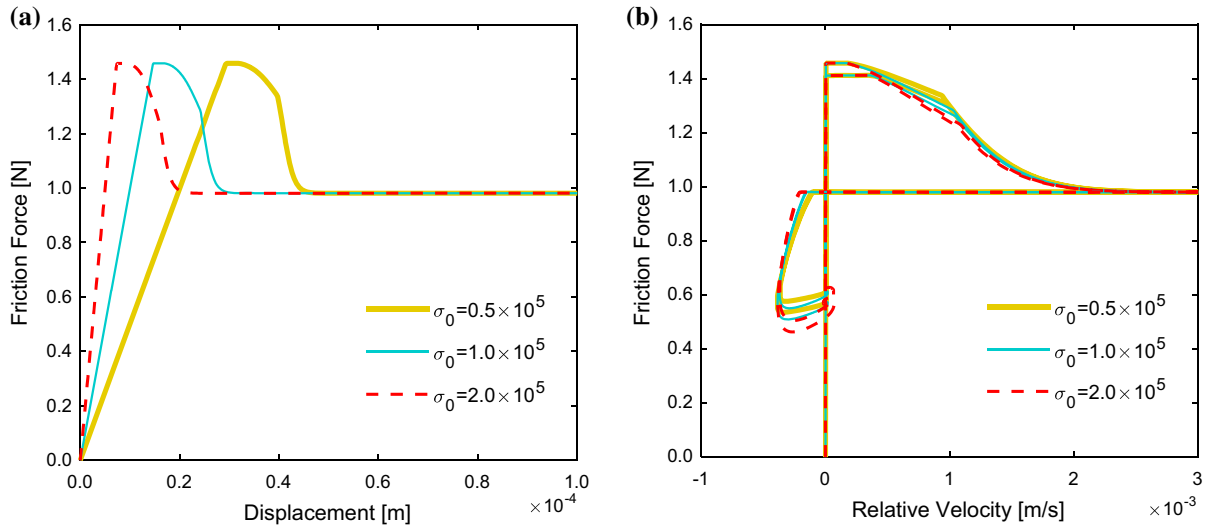


Fig. 30 Influence of stiffness coefficient (σ_0) in Gonthier model: **a** friction force versus displacement; **b** friction force versus relative velocity

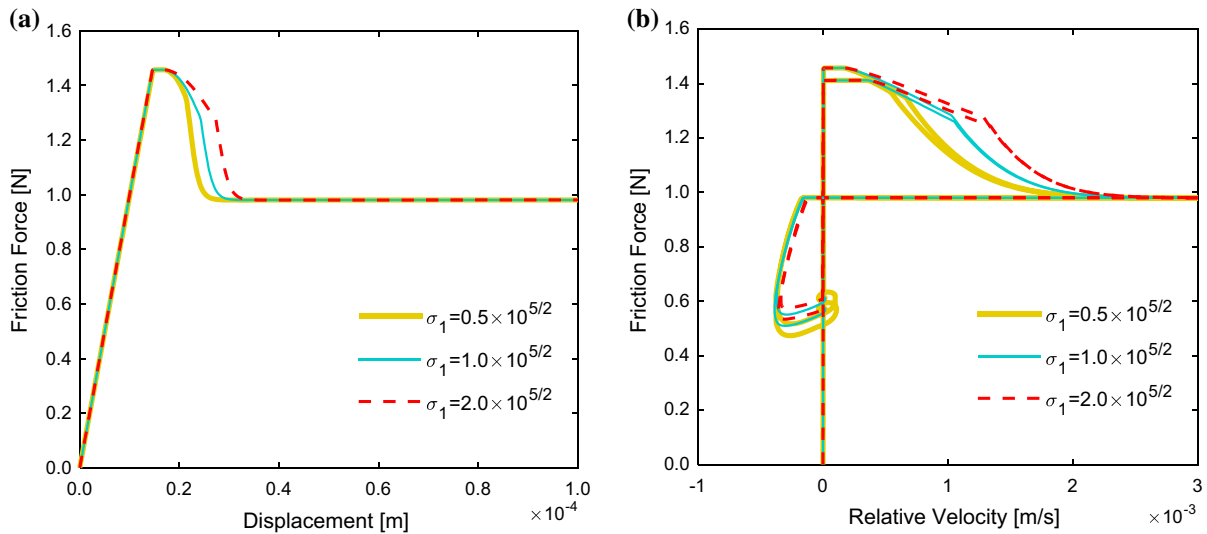


Fig. 31 Influence of damping coefficient (σ_1) in Gonthier model: **a** friction force versus displacement; **b** friction force versus relative velocity

generalities of the inclusion of friction in the multibody dynamics equations of motion are described next.

The equations of motion for a dynamic multibody system with holonomic constraints are given by the assembly of the Newton–Euler equations for a constrained multibody system and the accelerations constraint equations, which results in the following system [95]

$$\begin{bmatrix} \mathbf{M} & \mathbf{D}^T \\ \mathbf{D} & \mathbf{0} \end{bmatrix} \begin{Bmatrix} \dot{\mathbf{v}} \\ \boldsymbol{\lambda} \end{Bmatrix} = \begin{Bmatrix} \mathbf{g} \\ \boldsymbol{\gamma} \end{Bmatrix} \quad (73)$$

with the reference frame placed at the center of mass for each body, $\mathbf{M}$ is the system mass matrix, $\mathbf{D}$ is the Jacobian matrix of constraint equations, the vector $\dot{\mathbf{v}}$ contains the generalized state accelerations, $\boldsymbol{\lambda}$ is the vector that contains the Lagrange multipliers which represent

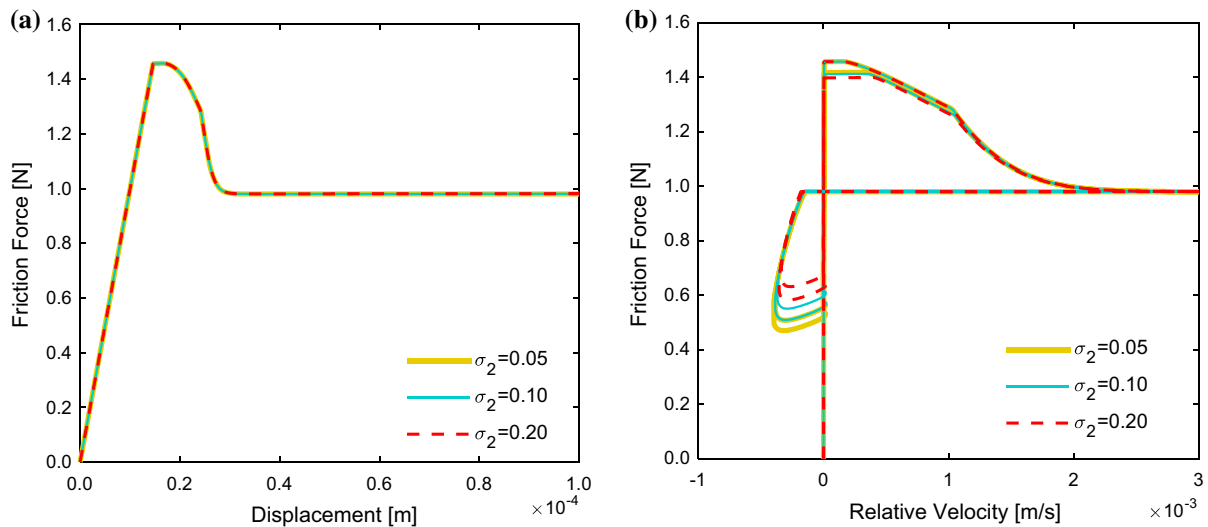


Fig. 32 Influence of viscosity coefficient (σ_2) in Gonthier model: **a** friction force versus displacement; **b** friction force versus relative velocity

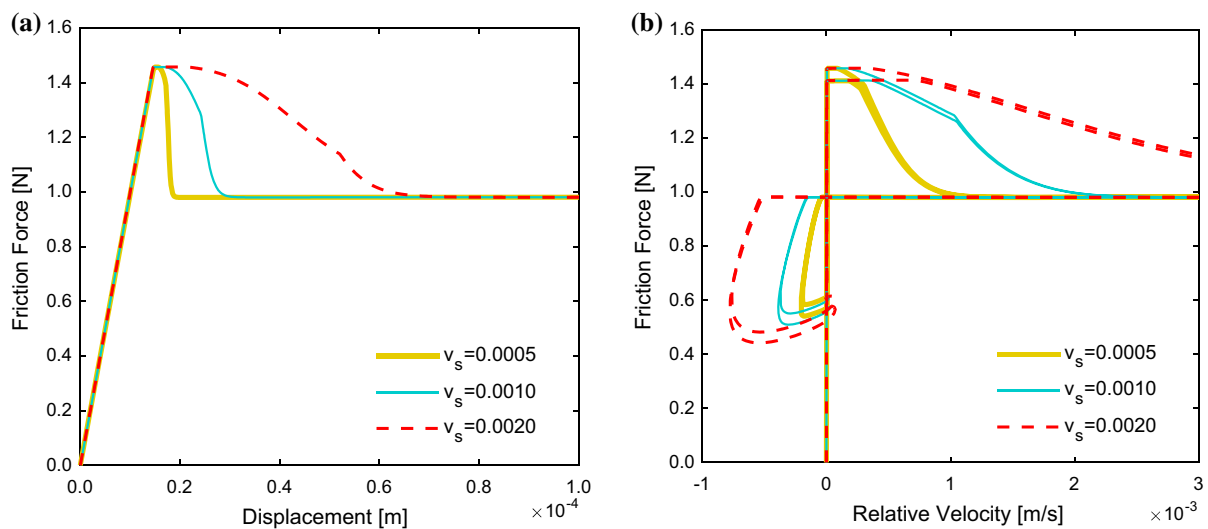


Fig. 33 Influence of Stribeck velocity (v_s) in Gonthier model: **a** friction force versus displacement; **b** friction force versus relative velocity

the reaction forces, $\mathbf{g}$ is the vector of generalized forces and $\mathbf{Y}$ is the vector of quadratic velocity terms.

Since the reaction forces are only calculated during the process of solving the problem's dynamics, and the friction forces are function of the joints' reactions, the implemented approach consists in including implicitly the calculation of the friction forces in the equations of motion [96]. Most of the friction models consider that the friction force is proportional to the magnitude of

the normal load; therefore, the first line of Eq. (73) can be adapted as

$$\mathbf{M}\dot{\mathbf{v}} + \mathbf{D}^T \boldsymbol{\lambda} + \mathbf{B} |\boldsymbol{\lambda}| = \mathbf{g} \quad (74)$$

where $\mathbf{B}$ is a matrix that establishes the relation of proportionality between the reaction and friction forces. However, Eq. (74) cannot be directly employed, so it can should be rewritten in the following form

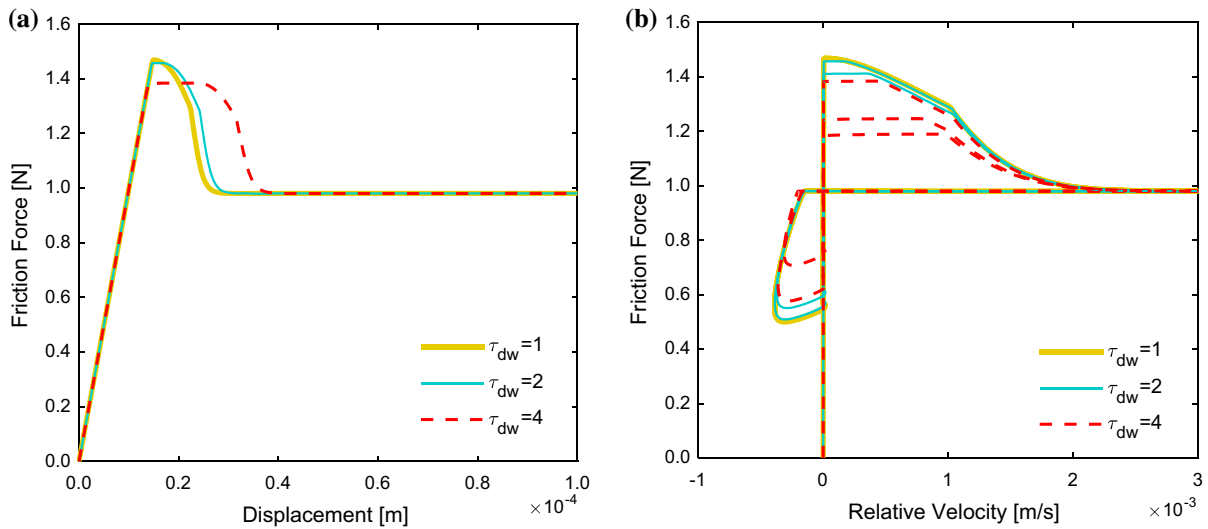


Fig. 34 Influence of dwell-time constant (τ_{dw}) in Gonthier model: **a** friction force versus displacement; **b** friction force versus relative velocity

Table 4 Impact of the friction parameters on the computing time

	<i>ode45</i>				<i>ode15s</i>			
	Dahl	Reset inte- grator	LuGre	Gonthier et al.	Dahl	Reset inte- grator	LuGre	Gonthier et al.
Curve shape parameter (α)	-19.73				-134.95			
Stiffness coefficient (σ_0)	1.48	0.63	1.51	0.82	0.79	0.06	0.32	-0.01
Damping coefficient (σ_1)		-0.08	0.00	-0.39		-0.47	-0.29	-0.21
Viscous coefficient (σ_2)			-0.06	0.01			-0.03	0.01
Stribeck velocity (v_S)			0.04	0.01			0.07	-0.05
Dwell-time constant τ_{dw}				0.00				0.18

$$\mathbf{M}\dot{\mathbf{v}} + \mathbf{D}^T \boldsymbol{\lambda} + \mathbf{B} \text{diag}(\text{sgn}(\boldsymbol{\lambda})) \boldsymbol{\lambda} = \mathbf{g} \quad (75)$$

where $\text{diag}(\mathbf{v})$ returns a square diagonal matrix with the elements of vector $\mathbf{v}$ on the main diagonal.

In order to solve the system dynamics with Eq. (75), it is necessary to evaluate the sign of the reaction forces which is unknown. To work around this problem, it is common to consider the sign of the reaction forces to be the same as the one from the previous time step. The drawback of this technique is the use of incorrect values in the steps that the reaction

forces change their sign. However, in these situations, the reaction forces are close to zero and, since they are proportional to the friction forces, the resultant error can be neglected. Thus, the matrix $\mathbf{B}$ can be rectified as

$$\mathbf{B}' = \mathbf{B} \text{diag}(\text{sgn}(\lambda_{i-1})) \quad (76)$$

Considering Eqs. (73)–(76), the equations of motion for constrained multibody systems with frictional constraints can be stated as

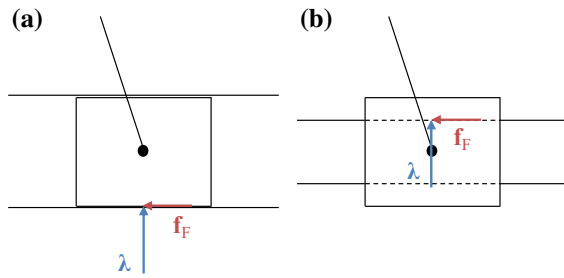


Fig. 35 Distinction of friction force acting on **a** an inner slider and **b** an outer slider

$$\begin{bmatrix} \mathbf{M} \mathbf{D}^T + \mathbf{B}' \\ \mathbf{D} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \dot{\mathbf{v}} \\ \dot{\lambda} \end{bmatrix} = \begin{bmatrix} \mathbf{g} \\ \gamma \end{bmatrix} \quad (77)$$

It should be noticed that, for the modeling of viscous friction forces, those components must be introduced in the equations of motion as external generalized forces, since they do not depend on the normal contact forces.

In order to correctly model the forces acting on the slider, it is necessary to know the details of its geometry. For an ideal joint, the dimensions of the slider are expendable, while, for this case, they must be defined since the friction force can create moment over the slider. Another important aspect consists of taking into account the type of the slider, which can be an inner or outer slider. This will influence the direction of the torque generated by the friction force, as represented in Fig. 35.

A spatial slider-crank mechanism is employed to demonstrate the effects of friction modeling in a translational joint, as represented in Fig. 36. The geometry and the inertial properties of each body in the system are presented in Table 5. The initial positions and velocities of the system are given in Table 6. For all the simulations, an inner slider with a cubic shape and a side length of 0.01 m was considered. Beyond the inertia and reaction forces, only gravitational forces are taken into account, which are applied in the negative direction of z axis as external forces.

The dynamic simulations of this mechanical system were carried for the frictionless case, and with eight of the most utilized friction models. The parameters for the friction force models are listed in Table 7. The final time of the simulations is 1 s, and the integration process was held with a fourth-order Runge–Kutta integrator with a time step equal to 1×10^{-5} s.

The results are analyzed through the motion of the slider corresponding to the simulations with the dif-

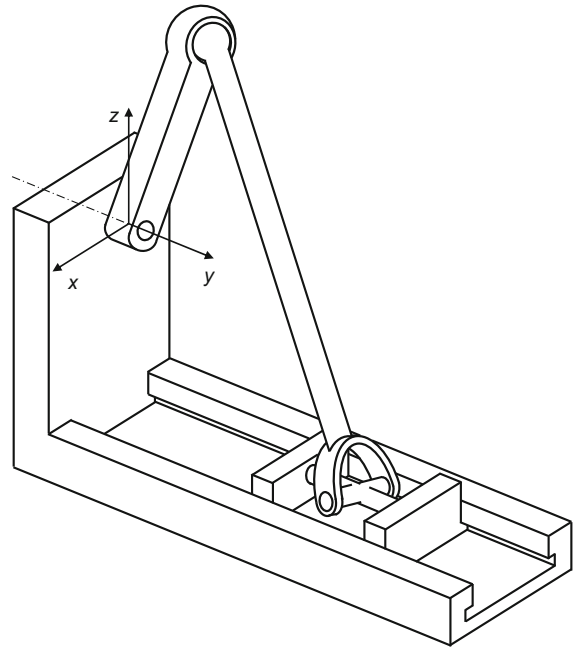


Fig. 36 Spatial slider-crank mechanism

Table 5 Geometrical and inertia properties of each body

Body	Length [m]	Mass [kg]	Principal moments of inertia [kg m ²]		
			$I_{\xi\xi}$	$I_{\eta\eta}$	$I_{\zeta\zeta}$
Crank	0.10	0.12	0.0001	0.0001	0.00001
Rod	0.29	0.5	0.004	0.0004	0.004
Slider	–	0.5	0.0001	0.0001	0.0001

ferent friction models, namely the plots for the position, velocity and acceleration. Similar to the previous example, for an easier comparison, the models were divided into two groups, models without and with stiction, which are displayed in Figs. 37 and 38, respectively. From the examination of position and velocity, it can be stated that the inclusion of friction modeling introduces an important dissipative effect, regardless of the frictional approach utilized. Figures 37c and 38c show the acceleration of the slider, which demonstrates more significant differences between the friction models, especially when the velocity changes direction. For this reason, a detailed analysis of the slider acceleration during the first time that the velocity changes its direction is represented in Figures 37d and 38d. In these plots, the models without stiction exhibit a rapid change

Table 6 Initial conditions for the dynamic analysis

Body	Initial positions						
	x	y	z	e_0	e_1	e_2	e_3
Crank	0.0000	0.0000	0.0500	1.0000	0.0000	0.0000	0.0000
Rod	0.0000	0.0735	−0.0250	0.8680	−0.4966	0.0000	0.0000
Slider	0.0000	0.1470	−0.1500	1.0000	0.0000	0.0000	0.0000
	Initial velocities						
	v_x	v_y	v_z	ω_x	ω_y	ω_z	
Crank	−0.3142	0.0000	0.0000	0.0000	−6.2832	0.0000	
Rod	−0.3142	0.0000	0.0000	0.0000	−1.5661	−1.6112	
Slider	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	

Table 7 Parameters considered for the different friction models

Parameter	Symbol	Value
Static coefficient of friction	μ_s	0.2
Kinetic coefficient of friction	μ_k	0.1
Stribeck velocity	v_s	0.001 m/s
Geometry factor	δ_σ	2
Tolerance velocity (Threlfall, Bengisu and Akay)	v_0	0.001 m/s
Tolerance velocity (Ambrósio)	v_0	0.0001 m/s
Tolerance velocity (Ambrósio)	v_1	0.001 m/s
Stiffness coefficient	σ_0	10^5 N/m
Damping coefficient	σ_1	$\sqrt{10^5}$ Ns/m
Coefficient of viscosity	σ_2	0 Ns/m

in the acceleration, which is reduced in one stage. Contrariwise, the models with stiction present three stages, the acceleration starts by increasing, then it has a bigger decrease, and finally, it slightly increases again. This behavior is more evident in the plots of Fig. 39, which represent the phase portraits of velocity vs. acceleration for the first change in velocity direction. These graphs can be used to analyze the relation between the friction force and velocity, since the variations of acceleration for a short period of time are highly associated with the variations in the friction force. From Fig. 39, one can state that the curves of static models, either with or

without stiction, present similar shape compared to the ones presented in Sections 2 and 3. Comparing with the model with static friction, it can be concluded that the dynamic ones also present the advantage of describing the frictional lag behavior, since they have a much higher friction force when the velocity is increasing than when it is decreasing.

For this example, the computational efficiency has also been assessed, but only with one integration algorithm (*ode15s*). The comparison of the computation time was achieved using the normalized computing time obtained with Eq. (71). The results are given in Table 8 and closely resemble the ones for the first example, in which the dynamic models require more effort than the static ones. In this case, one can compare the computational cost with the frictionless case, and it can be concluded that the inclusion of friction forces slows down the simulation.

This example allows for the examination of the effect of friction modeling in the dynamic analysis of a multi-body system, as well as the main differences between some of the most common friction force models.

5 Concluding remarks

A comparative study of some of the most relevant friction force models for multibody systems dynamics has been presented in this work. In the sequel of this process, the main characteristics, advantages and limitations of the static and dynamic friction force models were also analyzed. The most relevant issues

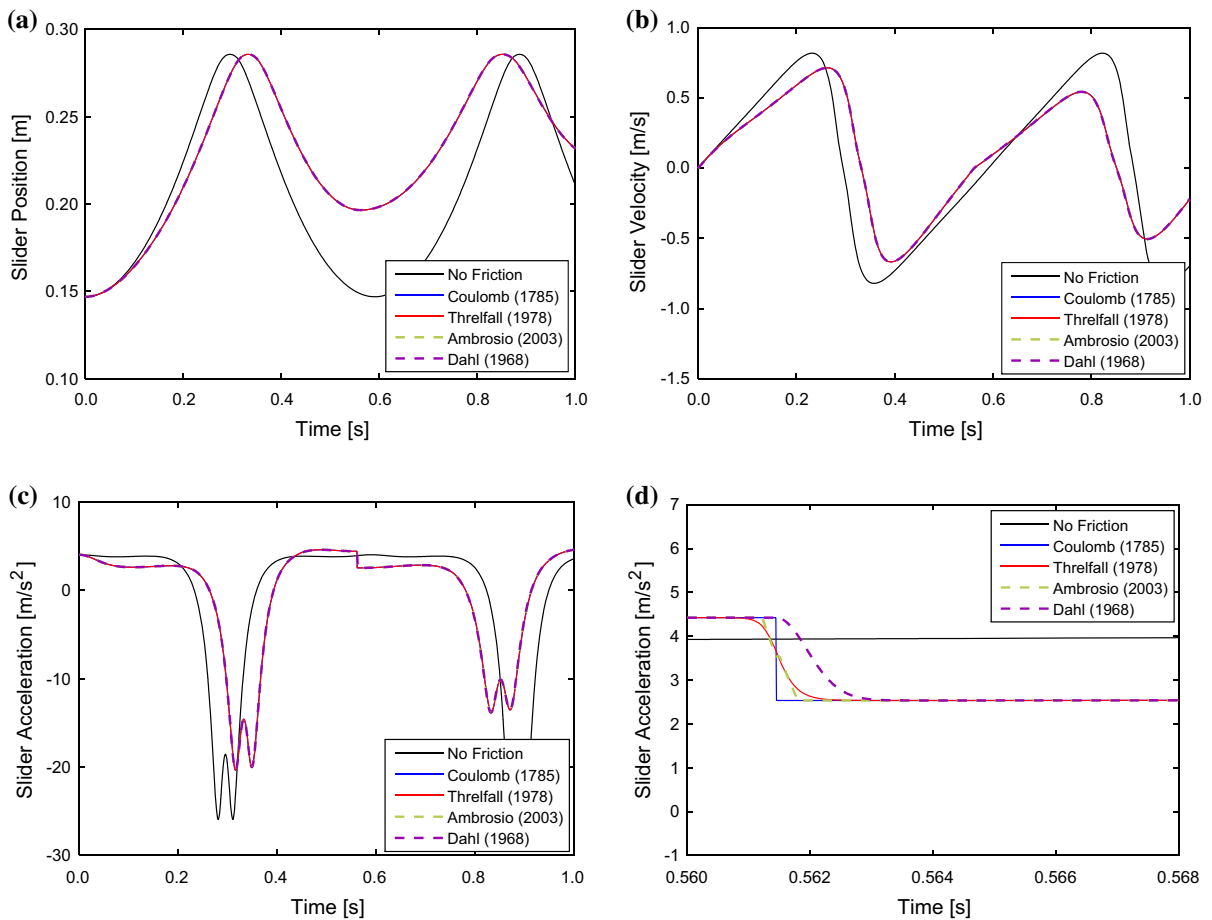


Fig. 37 Comparison of the motion of the slider for different friction models without stiction: **a** position; **b** velocity; **c** acceleration; **d** details of the acceleration plots

related to friction were addressed in order to understand the physical characteristics that the friction models should capture. The stick–slip motion, viscous friction, frictional lag, pre-sliding displacement and break-away force were among the discussed topics. On the one hand, the static friction models are, in general, simpler and describe the steady-state behavior of the friction force. Most of them though show incapacity to correctly capture the frictional effects. Several static models present a discontinuity of friction force at zero velocity, and this difficulty might lead to numerical instability during a dynamic simulation. On the other hand, the dynamic models use extra state variables to be able to take additional physical frictional phenomena into account. These approaches tend to be more complex and require, generally, the determination of more parameters.

To perform a comparative study of the dynamic response of the discussed models, a benchmark 1-DOF sliding mass–spring model was utilized first as an application example. This example showed that the most significant difference between friction models is the ability to capture the stick–slip motion. However, the viscous friction, the frictional lag or the pre-sliding displacement also has a relevant effect on the system dynamics. The inclusion of friction in a kinematic translational joint in a multibody system was also investigated. This analysis was carried with the dynamic study of spatial slider-crank mechanism. The results showed that friction introduces a high energy dissipation component, which has a pronounced effect on the velocity of the bodies.

Overall, the friction models present more significant differences at the accelerations level, mainly

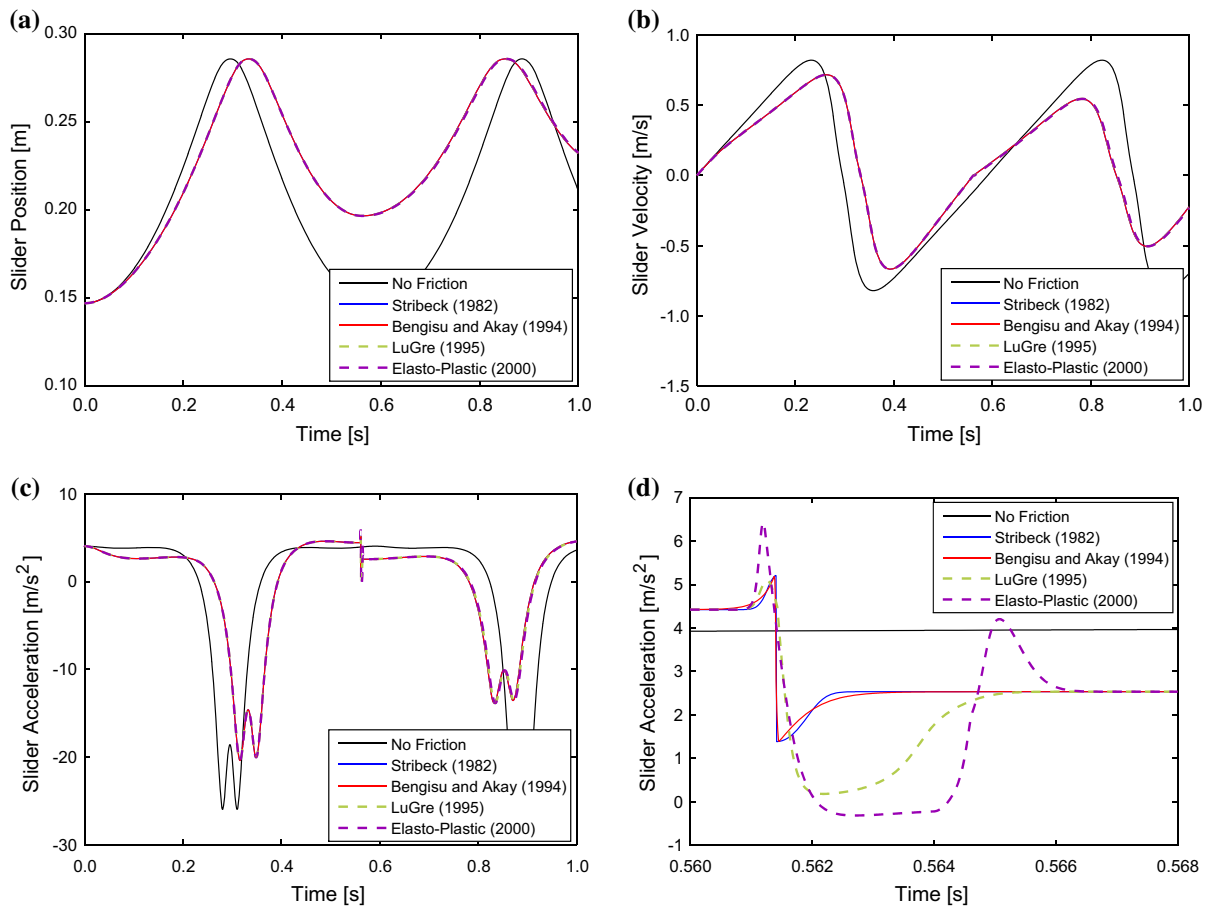


Fig. 38 Comparison of the motion of the slider for different friction models with stiction: **a** position; **b** velocity; **c** acceleration; **d** details of the acceleration plots

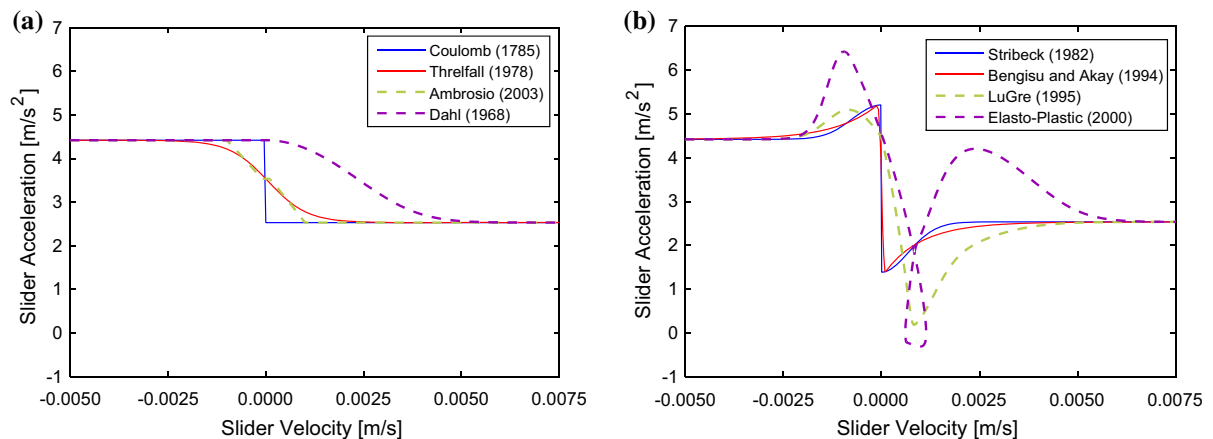


Fig. 39 Phase portrait velocity versus acceleration for the first change in velocity direction: **a** model without stiction; **b** models with stiction

Table 8 Normalized computing time ψ for the translational joint with friction

Model	<i>ode15s</i>
Frictionless	1.00
Coulomb	1.58
Threlfall	1.32
Ambrósio	1.53
Dahl	3.46
Stribeck	1.36
Bengisu & Akay	1.30
LuGre	3.65
Elasto-plastic	4.34

when there are changes in the velocity direction. Since the main differences between friction models occur for small relative velocities, the numerical integration scheme and the time step should be carefully selected in order to correctly capture those differences. The analysis of computational effort indicated that the modeling and inclusion of friction forces typically makes the problems normally quite stiff. Moreover, the utilization of dynamic models requires in general more time of calculation.

Generally, in order to have a detailed modeling of frictional phenomena, it is necessary to use a friction model which require a large number of parameters. In that sense, there exists a trade-off between the easiness to implement and the captured phenomena. In most cases, the determination of the parameters needs to be done by experimental work; otherwise, the parameters are chosen empirically. From the results of this present study, it was observed that some parameters maintain their influence independently of the friction model. However, few of them, such as the damping coefficient, have different effects based on the utilized friction force model. In general, the stiffness coefficient is related to the slope of the pre-sliding displacement curve. The damping coefficient diminishes the friction-velocity oscillations during the sticking phase, although when the magnitude of velocity is increasing, it can extend or decrease the static friction range. Furthermore, the viscosity coefficient does not produce a significant influence in the presented results. The increase in Stribeck velocity typically enlarges the extent of static friction. The parameters in general, however, produce mainly localized effects on the resulting friction forces not their steady-state values.

This study illustrates that in order to correctly model, analyze and simulate frictional behavior in multi-body systems, an appropriate friction model must be adopted. For the multibody dynamics community, the use of static friction models is probably the easy choice, since they require less computational time and a smaller number of selected parameters. However, when the nature of energy dissipation and its quantification is of importance, then the dynamic models are recommended.

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